2. Let $A, E \in \mathbb{C}^{n\times n}$. We have

$$\begin{aligned}\sigma(A+E) &\subseteq W(A+E) \subseteq W(A)+W(E)\\ &\subseteq \{a+b\in\mathbb{C} : a\in W(A), \quad b\in\mathbb{C} \quad \text{with} \quad |b|\le \|E\|_2\}.\end{aligned}$$

3. Let $A \in \mathbb{C}^{n\times n}$ and $a \in \mathbb{C}$. Then $a \in \sigma(A)\cap\partial W(A)$ if and only if A is unitarily similar to $aI_k \oplus B$ such that $a \notin \sigma(B)\cup \operatorname{int}(W(B))$.
4. Let $A \in \mathbb{C}^{n\times n}$ and $a \in \mathbb{C}$. Then a is a nondifferentiable boundary point of $W(A)$ if and only if A is unitarily similar to $aI_k \oplus B$ such that $a \notin W(B)$. In particular, a is a reducing eigenvalue of A.
5. Let $A \in \mathbb{C}^{n\times n}$. If $W(A)$ has at least $n-1$ nondifferentiable boundary points or if at least $n-1$ eigenvalues of A (counting multiplicities) lie in $\partial W(A)$, then A is normal.

Examples:

1. Let $A = [1] \oplus \begin{bmatrix} 0 & 2\\ 0 & 0\end{bmatrix}$. Then $W(A)$ is the unit disk centered at the origin, and 1 is a reducing eigenvalue of A lying on the boundary of $W(A)$.
2. Let $A = [2] \oplus \begin{bmatrix} 0 & 2\\ 0 & 0\end{bmatrix}$. Then $W(A)$ is the convex hull of unit disk centered at the origin of the number 2, and 2 is a nondifferentiable boundary point of $W(A)$.

Applications:

1. By Fact 1, if $A \in \mathbb{C}^{n\times n}$ and $0 \notin W(A)$, then $0 \notin \sigma(A)$ and, thus, A is invertible.
2. By Fact 4, if $A \in \mathbb{C}^{n\times n}$, then $W(A)$ has at most n nondifferentiable boundary points.
3. While $W(A)$ does not give a very tight containment region for $\sigma(A)$ as shown in the examples in the last section. Fact 2 shows that the numerical range can be used to estimated the spectrum of the resulting matrix when A is under a perturbation E. In contrast, $\sigma(A)$ and $\sigma(E)$ usually do not carry much information about $\sigma(A+E)$ in general. For example, let $A = \begin{bmatrix} 0 & M\\ 0 & 0\end{bmatrix}$ and $E = \begin{bmatrix} 0 & 0\\ \varepsilon & 0\end{bmatrix}$. Then $\sigma(A) = \sigma(E) = \{0\}$, $\sigma(A+E) = \{\pm\sqrt{M\varepsilon}\} \subseteq W(A+E)$, which is the elliptical disk with foci $\pm\sqrt{M\varepsilon}$ and length of minor axis equal to $|\,|M|-|\varepsilon|\,|$.

18.3 Location of the Numerical Range

Facts:

The following facts can be found in [HJ91].

1. Let $A \in \mathbb{C}^{n\times n}$ and $t \in [0, 2\pi)$. Suppose $\mathbf{x}_t \in \mathbb{C}^n$ is a unit eigenvector corresponding to the largest eigenvalue $\lambda_1(t)$ of $e^{it}A + e^{-it}A^*$, and

$$\mathcal{P}_t = \{a \in \mathbb{C} : e^{it}a + e^{-it}\bar{a} \le \lambda_1(t)\}.$$

Then

$$e^{it}W(A) \subseteq \mathcal{P}_t, \quad \lambda_t = \mathbf{x}_t^* A\mathbf{x}_t \in \partial W(A)\cap\partial\mathcal{P}_t$$

and

$$W(A) = \cap_{r\in[0,2\pi)}\, e^{-ir}\mathcal{P}_r = \text{conv}\,\{\lambda_r : r \in [0, 2\pi)\}.$$

If $T = \{t_1, \ldots, t_k\}$ with $0 \leq t_1 < \cdots < t_k < 2\pi$ and $k > 2$ such that $t_k - t_1 > \pi$, then

$$P_T^O(A) = \cap_{r\in T}\, e^{-ir} P_r \quad \text{and} \quad P_T^I(A) = \text{conv}\,\{\lambda_r : r \in T\}$$

are two polygons in $\mathbb{C}$ such that

$$P_T^I(A) \subseteq W(A) \subseteq P_T^O(A).$$

Moreover, both the area $W(A)\backslash P_T^I(A)$ and the area of $P_T^O(A)\backslash W(A)$ converge to 0 as $\max\{t_j - t_{j-1} : 1 \leq j \leq k+1\}$ converges to 0, where $t_0 = 0, t_{k+1} = 2\pi$.

2. Let $A = (a_{ij}) \in \mathbb{C}^{n\times n}$. For each $j = 1, \ldots, n$, let

$$g_j = \sum_{i\neq j}(|a_{ij}| + |a_{ji}|)/2 \quad \text{and} \quad G_j(A) = \{a \in \mathbb{C} : |a - a_{jj}| \leq g_j\}.$$

Then

$$W(A) \subseteq \text{conv}\,\cup_{j=1}^n G_j(A).$$

Examples:

1. Let $A = \begin{bmatrix} 2 & 2 \\ 0 & 2 \end{bmatrix}$. Then $W(A)$ is the circular disk centered at 2 with radius 1 In Figure 18.2, $W(A)$ is approximated by $P_T^O(A)$ with $T = \{2k\pi/100 : 0 \leq k \leq 99\}$. If $T = \{0, \pi/2, \pi, 3\pi/2\}$, then the polygon $P_T^O(A)$ in Fact 1 is bounded by the four lines $\{3 + bi : b \in \mathbb{R}\}$, $\{a + i : a \in \mathbb{R}\}$, $\{1 + bi : b \in \mathbb{R}\}$, $\{a - i : a \in \mathbb{R}\}$, and the polygon $P_T^I(A)$ equals the convex hull of $\{2, 1 + i, 0, 1 - i\}$.
2. Let $A = \begin{bmatrix} 5i & 2 & 3 \\ 4 & -3i & -2 \\ 1 & 3 & 9 \end{bmatrix}$. In Figure 18.3, $W(A)$ is approximated by $P_T^O(A)$ with $T = \{2k\pi/100 : 0 \leq k \leq 99\}$. By Fact 2, $W(A)$ lies in the convex hull of the circles $G_1 = \{a \in \mathbb{C} : |a - 5i| \leq 5\}$, $G_2 = \{a \in \mathbb{C} : |a + 3i| \leq 5.5\}$, $G_3 = \{a \in \mathbb{C} : |a - 9| \leq 4.5\}$.

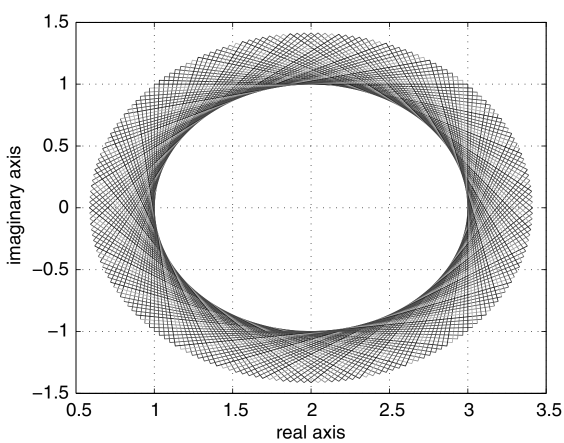

FIGURE 18.2 Numerical range of the matrix A in Example 1.

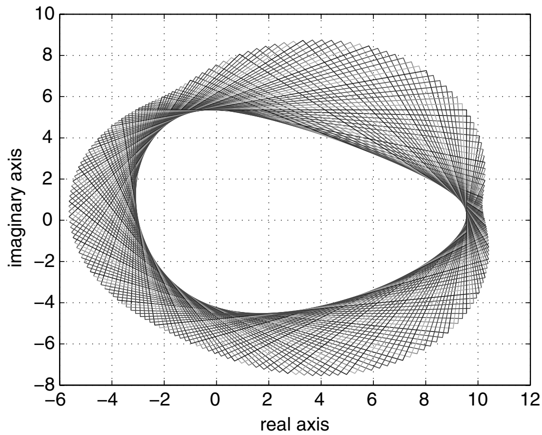

FIGURE 18.3 Numerical range of the matrix A in Example 2.

Applications:

1. Let $A = H + iG$, where $H, G \in \mathbb{C}^{n\times n}$ are Hermitian. Then

$$W(A) \subseteq W(H) + iW(G) = \{a + ib : a \in W(H),\ b \in W(G)\},$$

which is $P_T^O(A)$ for $T = \{0, \pi/2, \pi, 3\pi/2\}$.

2. Let $A = H + iG$, where $H, G \in \mathbb{C}^{n\times n}$ are Hermitian. Denote by $\lambda_1(X) \geq \cdots \geq \lambda_n(X)$ for a Hermitian matrix $X \in \mathbb{C}^{n\times n}$. By Fact 1,

$$w(A) = \max\{\lambda_1(\cos tH + \sin tG) : t \in [0, 2\pi)\}.$$

If $0 \notin W(A)$, then

$$\widetilde{w}(A) = \max\{\{\lambda_n(\cos tH + \sin tG) : t \in [0, 2\pi)\} \cup \{0\}\}.$$

3. By Fact 2, if $A = (a_{ij}) \in \mathbb{C}^{n\times n}$, then

$$w(A) \leq \max\{|a_{jj}| + g_j : 1 \leq j \leq n\}.$$

In particular, if A is nonnegative, then $w(A) = \lambda_1(A + A^T)/2$.

18.4 Numerical Radius

Definitions:

Let N be a vector norm on $\mathbb{C}^{n\times n}$. It is **submultiplicative** if

$$N(AB) \leq N(A)N(B) \quad \text{for all} \quad A, B \in \mathbb{C}^{n\times n}.$$

It is **unitarily invariant** if

$$N(UAV) = N(A) \quad \text{for all} \quad A \in \mathbb{C}^{n\times n} \text{ and unitary } U, V \in \mathbb{C}^{n\times n}.$$

It is **unitary similarity invariant** (also known as **weakly unitarily invariant**) if

$$N(U^*AU) = N(A) \quad \text{for all} \quad A \in \mathbb{C}^{n\times n} \text{ and unitary } U \in \mathbb{C}^{n\times n}.$$

Facts:

The following facts can be found in [GR96] and [HJ91].

1. The numerical radius $w(\cdot)$ is a unitary similarity invariant vector norm on $\mathbb{C}^{n\times n}$, and it is not unitarily invariant.
2. For any $A \in \mathbb{C}^{n\times n}$, we have
$$\rho(A) \leq w(A) \leq \|A\|_2 \leq 2w(A).$$
3. Suppose $A \in \mathbb{C}^{n\times n}$ is nonzero and the minimal polynomial of A has degree m. The following conditions are equivalent.
   (a) $\rho(A) = w(A)$.
   (b) There exists $k \geq 1$ such that A is unitarily similar to $\gamma U \oplus B$ for a unitary $U \in \mathbb{C}^{k\times k}$ and $B \in \mathbb{C}^{(n-k)\times(n-k)}$ with $w(B) \leq w(A) = \gamma$.
   (c) There exists $s \geq m$ such that $w(A^s) = w(A)^s$.
4. Suppose $A \in \mathbb{C}^{n\times n}$ is nonzero and the minimal polynomial of A has degree m. The following conditions are equivalent.
   (a) $\rho(A) = \|A\|_2$.
   (b) $w(A) = \|A\|_2$.
   (c) There exists $k \geq 1$ such that A is unitarily similar to $\gamma U \oplus B$ for a unitary $U \in \mathbb{C}^{k\times k}$ and a $B \in \mathbb{C}^{(n-k)\times(n-k)}$ with $\|B\|_2 \leq \|A\|_2 = \gamma$.
   (d) There exists $s \geq m$ such that $\|A^s\|_2 = \|A\|_2^s$.
5. Suppose $A \in \mathbb{C}^{n\times n}$ is nonzero. The following conditions are equivalent.
   (a) $\|A\|_2 = 2w(A)$.
   (b) $W(A)$ is a circular disk centered at origin with radius $\|A\|_2/2$.
   (c) $A/\|A\|_2$ is unitarily similar to $A_1 \oplus A_2$ such that $A_1 = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix}$ and $w(A_2) \leq 1$.
6. The vector norm $4w$ on $\mathbb{C}^{n\times n}$ is submultiplicative, i.e.,
$$4w(AB) \leq (4w(A))(4w(B)) \quad \text{for all} \quad A, B \in \mathbb{C}^{n\times n}.$$
   The equality holds if
$$X = Y^T = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix}.$$
7. Let $A \in \mathbb{C}^{n\times n}$ and k be a positive integer. Then
$$w(A^k) \leq w(A)^k.$$
8. Let N be a unitary similarity invariant vector norm on $\mathbb{C}^{n\times n}$ such that $N(A^k) \leq N(A)^k$ for any $A \in \mathbb{C}^{n\times n}$ and positive integer k. Then
$$w(A) \leq N(A) \quad \text{for all} \quad A \in \mathbb{C}^{n\times n}.$$
9. Suppose N is a unitarily invariant vector norm on $\mathbb{C}^{n\times n}$. Let
$$D = \begin{cases} 2I_k \oplus 0_k & \text{if } n = 2k, \\ 2I_k \oplus I_1 \oplus 0_k & \text{if } n = 2k+1. \end{cases}$$

Then $a = N(E_{11})$ and $b = N(D)$ are the best (largest and smallest) constants such that

$$aw(A) \le N(A) \le bw(A) \quad \text{for all} \quad A \in \mathbb{C}^{n\times n}.$$

10. Let $A \in \mathbb{C}^{n\times n}$. The following are equivalent:
    (a) $w(A) \le 1$.
    (b) $\lambda_1(e^{it}A + e^{-it}A^*)/2 \le 1$ for all $t \in [0, 2\pi)$.
    (c) There is $Z \in \mathbb{C}^{n\times n}$ such that $\begin{bmatrix} I_n + Z & A \\ A^* & I_n - Z \end{bmatrix}$ is positive semidefinite.
    (d) There exists $X \in \mathbb{C}^{2n\times n}$ satisfying $X^*X = I_n$ and
$$A = X^* \begin{bmatrix} 0_n & 2I_n \\ 0_n & 0_n \end{bmatrix} X.$$

18.5 Products of Matrices

Facts:

The following facts can be found in [GR96] and [HJ91].

1. Let $A, B \in \mathbb{C}^{n\times n}$ be such that $\widetilde{w}(A) > 0$. Then
$$\sigma(A^{-1}B) \subseteq \{b/a : a \in W(A), b \in W(B)\}.$$
2. Let $0 \le t_1 < t_2 < t_1 + \pi$ and $S = \{re^{it} : r > 0,\ t \in [t_1, t_2]\}$. Then $\sigma(A) \subseteq S$ if and only if there is a positive definite $B \in \mathbb{C}^{n\times n}$ such that $W(AB) \subseteq S$.
3. Let $A, B \in \mathbb{C}^{n\times n}$.
    (a) If $AB = BA$, then $w(AB) \le 2w(A)w(B)$.
    (b) If A or B is normal such that $AB = BA$, then $w(AB) \le w(A)w(B)$.
    (c) If $A^2 = aI$ and $AB = BA$, then $w(AB) \le \|A\|_2 w(B)$.
    (d) If $AB = BA$ and $AB^* = B^*A$, then $w(AB) \le \min\{w(A)\|B\|_2, \|A\|_2 w(B)\}$.
4. Let A and B be square matrices such that A or B is normal. Then
$$W(A \circ B) \subseteq W(A \otimes B) = \operatorname{conv}\{W(A)W(B)\}.$$
Consequently,
$$w(A \circ B) \le w(A \otimes B) = w(A)w(B).$$
(See Chapter 8.5 and 10.4 for the definitions of t $A \circ B$ and $A \otimes B$.)
5. Let A and B be square matrices. Then
$$w(A \circ B) \le w(A \otimes B) \le \min\{w(A)\|B\|_2, \|A\|_2 w(B)\} \le 2w(A)w(B).$$
6. Let $A \in \mathbb{C}^{n\times n}$. Then
$$w(A \circ X) \le w(X) \quad \text{for all} \quad X \in \mathbb{C}^{n\times n}$$
if and only if $A = B^*WB$ such that W satisfies $\|W\| \le 1$ and all diagonal entries of B^*B are bounded by 1.

Examples:

1. Let $A \in \mathbb{C}^{9\times 9}$ be the Jordan block of zero $J_9(0)$, and $B = A^3 + A^7$. Then $w(A) = w(B) = \cos(\pi/10) < 1$ and $w(AB) = 1 > \|A\|_2 w(B)$. So, even if $AB = BA$, we may not have $w(AB) \leq \min\{w(A)\|B\|_2, \|A\|_2 w(B)\}$.
2. Let $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. Then $W(A) = \{a \in \mathbb{C} : |a - 1| \leq 1/2\}$ and
$$W(A^2) = \{a \in \mathbb{C} : |a - 1| \leq 1\},$$
whereas
$$\operatorname{conv} W(A)^2 \subseteq \{se^{it} \in \mathbb{C} : s \in [0.25, 2.25],\ t \in [-\pi/3, \pi/3]\}.$$
So, $W(A^2) \not\subseteq \operatorname{conv} W(A)^2$.
3. Let $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. Then $\sigma(AB) = \{i, -i\}$, $W(AB) = i[-1, 1]$, and $W(A) = W(B) = W(A)W(B) = [-1, 1]$. So, $\sigma(AB) \not\subseteq \operatorname{conv} W(A)W(B)$.

Applications:

1. If $C \in \mathbb{C}^{n\times n}$ is positive definite, then $W(C^{-1}) = W(C)^{-1} = \{c^{-1} : c \in W(C)\}$. Applying Fact 1 with $A = C^{-1}$,
$$\sigma(CB) \subseteq W(C)W(B).$$
2. If $C \in \mathbb{C}^{n\times n}$ satisfies $\widetilde{w}(C) > 0$, then for every unit vector $\mathbf{x} \in \mathbb{C}^n$ $\mathbf{x}^* C^{-1}\mathbf{x} = \mathbf{y}^* C^* C^{-1} C\mathbf{y}$ with $\mathbf{y} = C^{-1}\mathbf{x}$ and, hence,
$$W(C^{-1}) \subseteq \{rb : r \geq 0, \quad b \in W(C^*)\} = \{r\overline{b} : r \geq 0, \quad b \in W(C)\}.$$
Applying this observation and Fact 1 with $A = C^{-1}$, we have
$$\sigma(AB) \subseteq \{rab : r \geq 0, \quad a \in W(A), \quad b \in W(B)\}.$$

18.6 Dilations and Norm Estimation

Definitions:

A matrix $A \in \mathbb{C}^{n\times n}$ has a **dilation** $B \in \mathbb{C}^{m\times m}$ if there is $X \in \mathbb{C}^{m\times n}$ such that $X^*X = I_n$ and $X^*BX = A$.

A matrix $A \in \mathbb{C}^{n\times n}$ is a **contraction** if $\|A\|_2 \leq 1$.

Facts:

The following facts can be found in [CL00],[CL01] and their references.

1. A has a dilation B if and only if B is unitarily similar to a matrix of the form
$$\begin{bmatrix} A & * \\ * & * \end{bmatrix}.$$
2. Suppose $B \in \mathbb{C}^{3\times 3}$ has a reducing eigenvalue, or $B \in \mathbb{C}^{2\times 2}$. If $W(A) \subseteq W(B)$, then A has a dilation of the form $B \otimes I_m$.

3. Let $r \in [-1, 1]$. Suppose $A \in \mathbb{C}^{n\times n}$ is a contraction with

$$W(A) \subseteq \mathcal{S} = \{a \in \mathbb{C} : a + \bar{a} \leq 2r\}.$$

Then A has a unitary dilation $U \in \mathbb{C}^{2n\times 2n}$ such that $W(U) \subseteq \mathcal{S}$.

4. Let $A \in \mathbb{C}^{n\times n}$. Then

$$W(A) = \cap\{W(B) : B \in \mathbb{C}^{2n\times 2n} \text{ is a normal dilation of } A\}.$$

If A is a contraction, then

$$W(A) = \cap\{W(U) : U \in \mathbb{C}^{2n\times 2n} \text{ is a unitary dilation of } A\}.$$

5. Let $A \in \mathbb{C}^{n\times n}$.
   (a) If $W(A)$ lies in an triangle with vertices z_1, z_2, z_3, then

$$\|A\|_2 \leq \max\{|z_1|, |z_2|, |z_3|\}.$$

   (b) If $W(A)$ lies in an ellipse $\mathcal{E}$ with foci λ_1, λ_2, and minor axis of length b, then

$$\|A\|_2 \leq \{\sqrt{(|\lambda_1| + |\lambda_2|)^2 + b^2} + \sqrt{(|\lambda_1| - |\lambda_2|)^2 + b^2}\}/2.$$

   More generally, if $W(A)$ lies in the convex hull of the ellipse $\mathcal{E}$ and the point z_0, then

$$\|A\|_2 \leq \max\left\{|z_0|, \{\sqrt{(|\lambda_1| + |\lambda_2|)^2 + b^2} + \sqrt{(|\lambda_1| - |\lambda_2|)^2 + b^2}\}/2\right\}.$$

6. Let $A \in \mathbb{C}^{n\times n}$. Suppose there is $t \in [0, 2\pi)$ such that $e^{it}W(A)$ lies in a rectangle R centered at $z_0 \in \mathbb{C}$ with vertices $z_0 \pm \alpha \pm i\beta$ and $z_0 \pm \alpha \mp i\beta$, where $\alpha, \beta > 0$, so that $z_1 = z_0 + \alpha + i\beta$ has the largest magnitude. Then

$$\|A\|_2 \leq \begin{cases} |z_1| & \text{if } R \subseteq \operatorname{conv}\{z_1, \bar{z}_1, -\bar{z}_1\}, \\ \alpha + \beta & \text{otherwise.} \end{cases}$$

The bound in each case is attainable.

Examples:

1. Let $A = \begin{bmatrix} 0 & \sqrt{2} \\ 0 & 0 \end{bmatrix}$. Suppose

$$B = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{or} \quad B = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & i & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -i \end{bmatrix}.$$

Then $W(A) \subseteq W(B)$. However, A does not have a dilation of the form $B \otimes I_m$ for either of the matrices because

$$\|A\|_2 = \sqrt{2} > 1 = \|B\|_2 = \|B \otimes I_m\|_2.$$

So, there is no hope to further extend Fact 1 in this section to arbitrary $B \in \mathbb{C}^{3\times 3}$ or normal matrix $B \in \mathbb{C}^{4\times 4}$.

18.7 Mappings on Matrices

Definitions:

Let $\phi : \mathbb{C}^{n\times n} \to \mathbb{C}^{m\times m}$ be a linear map. It is **unital** if $\phi(I_n) = I_m$; it is **positive** if $\phi(A)$ is positive semidefinite whenever A is positive semidefinite.

Facts:

The following facts can be found in [GR96] unless another reference is given.

1. [HJ91] Let $\mathcal{P}(\mathbb{C})$ be the set of subsets of $\mathbb{C}$. Suppose a function $F : \mathbb{C}^{n\times n} \to \mathcal{P}(\mathbb{C})$ satisfies the following three conditions.
   (a) $F(A)$ is compact and convex for every $A \in \mathbb{C}^{n\times n}$.
   (b) $F(aA + bI) = aF(A) + b$ for any $a, b \in \mathbb{C}$ and $A \in \mathbb{C}^{n\times n}$.
   (c) $F(A) \subseteq \{a \in \mathbb{C} : a + \bar{a} \geq 0\}$ if and only if $A + A^*$ is positive semidefinite.

   Then $F(A) = W(A)$ for all $A \in \mathbb{C}^{n\times n}$.
2. Use the usual topology on $\mathbb{C}^{n\times n}$ and the Hausdorff metric on two compact sets A, B of C defined by
$$d(\mathcal{A}, \mathcal{B}) = \max\left\{\max_{a\in\mathcal{A}} \quad \min_{b\in\mathcal{B}} |a-b|,\ \max_{b\in\mathcal{B}} \quad \min_{a\in\mathcal{A}} |a-b|\right\}$$
   The mapping $A \mapsto W(A)$ is continuous.
3. Suppose $f(x+iy) = (ax+by+c)+i(dx+ey+f)$ for some real numbers a, b, c, d, e, f. Define $f(H+iG) = (aH+bG+cI)+i(dH+eG+fI)$ for any two Hermitian matrices $H, G \in \mathbb{C}^{n\times n}$. We have
$$W(f(H+iG)) = f(W(A)) = \{f(x+iy) : x+iy \in W(A)\}.$$
4. Let $\mathbf{D} = \{a \in \mathbb{C} : |a| \leq 1\}$. Suppose $f : \mathbf{D} \to \mathbb{C}$ is analytic in the interior of $\mathbf{D}$ and continuous on the boundary of $\mathbf{D}$.
   (a) If $f(\mathbf{D}) \subseteq \mathbf{D}$ and $f(0) = 0$, then $W(f(A)) \subseteq \mathbf{D}$ whenever $W(A) \subseteq \mathbf{D}$.
   (b) If $f(\mathbf{D}) \subseteq \mathbb{C}_+ = \{a \in \mathbb{C} : a + \bar{a} \geq 0\}$, then $W(f(A)) \subseteq \mathbb{C}_+ \setminus \{(f(0) + \overline{f(0)})/2\}$ whenever $W(A) \subseteq \mathbf{D}$.
5. Suppose $\phi : \mathbb{C}^{n\times n} \to \mathbb{C}^{n\times n}$ is a unital positive linear map. Then $W(\phi(A)) \subseteq W(A)$ for all $A \in \mathbb{C}^{n\times n}$.
6. [Pel75] Let $\phi : \mathbb{C}^{n\times n} \to \mathbb{C}^{n\times n}$ be linear. Then
$$W(A) = W(\phi(A)) \quad \text{for all} \quad A \in \mathbb{C}^{n\times n}$$
   if and only if there is a unitary $U \in \mathbb{C}^{n\times n}$ such that ϕ has the form
$$X \mapsto U^*XU \quad \text{or} \quad X \mapsto U^*X^TU.$$
7. [Li87] Let $\phi : \mathbb{C}^{n\times n} \to \mathbb{C}^{n\times n}$ be linear. Then $w(A) = w(\phi(A))$ for all $A \in \mathbb{C}^{n\times n}$ if and only if there exist a unitary $U \in \mathbb{C}^{n\times n}$ and a complex unit μ such that ϕ has the form
$$X \mapsto \mu U^*XU \quad \text{or} \quad X \mapsto \mu U^*X^TU.$$

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19

Matrix Stability and Inertia

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Much is known about spectral properties of square (complex) matrices. There are extensive studies of eigenvalues of matrices in certain classes. Some of the studies concentrate on the inertia of the matrices, that is, distribution of the eigenvalues in half-planes.

A special inertia case is of stable matrices, that is, matrices whose spectrum lies in the open left or right half-plane. These, and other related types of matrix stability, play an important role in various applications. For this reason, matrix stability has been intensively investigated in the past two centuries.

A. M. Lyapunov, called by F. R. Gantmacher "the founder of the modern theory of stability," studied the asymptotic stability of solutions of differential systems. In 1892, he proved a theorem that was restated (first, apparently, by Gantmacher in 1953) as a necessary and sufficient condition for stability of a matrix. In 1875, E. J. Routh introduced an algorithm that provides a criterion for stability. An independent solution was given by A. Hurwitz. This solution is known nowadays as the Routh–Hurwitz criterion for stability. Another criterion for stability, which has a computational advantage over the Routh–Hurwitz criterion, was proved in 1914 by Liénard and Chipart. The equivalent of the Routh–Hurwitz and Liénard–Chipart criteria was observed by M. Fujiwara. The related problem of requiring the eigenvalues to be within the unit circle was solved separately in the early 1900s by I. Schur and Cohn. The above-mentioned studies have motivated an intensive search for conditions for matrix stability.

An interesting question, related to stability, is the following one: Given a square matrix A, can we find a diagonal matrix D such that the matrix DA is stable? This question can be asked in full generality, as suggested above, or with some restrictions on the matrix D, such as positivity of the diagonal elements. A related problem is characterizing matrices A such that for every positive diagonal matrix D, the matrix DA is stable. Such matrices are called multiplicative D-stable matrices. This type of matrix stability, as well as two other related types, namely additive D-stability and Lyapunov diagonal (semi)stability, have important applications in many disciplines. Thus, they are very important to characterize. While regular stability is a spectral property (it is always possible to check whether a given matrix is stable or not by evaluating its eigenvalues), none of the other three types of matrix stability can be characterized by the spectrum of the matrix. This problem has been solved for certain classes of matrices. For example, for Z-matrices all the stability types are equivalent. Another case in which these characterization problems have been solved is the case of acyclic matrices.

Several surveys handle the above-mentioned types of matrix stability, e.g., the books [HJ91] and [KB00], and the articles [Her92], [Her98], and [BH85]. Finally, the mathematical literature has studies of other types of matrix stability, e.g., the above-mentioned Schur–Cohn stability (where all the eigenvalues lie within the unit circle), e.g., [Sch17] and [Zah92]; H-stability, e.g., [OS62], [Car68], and [HM98]; L_2-stability and strict H-stability, e.g., [Tad81]; and scalar stability, e.g., [HM98].

19.1 Inertia

Much is known about spectral properties of square matrices. In this chapter, we concentrate on the distribution of the eigenvalues in half-planes. In particular, we refer to results that involve the expression $AH + HA^*$, where A is a square complex matrix and H is a Hermitian matrix.

Definitions:

For a square complex matrix A, we denote by $\pi(A)$ the number of eigenvalues of A with positive real part, by $\delta(A)$ the number of eigenvalues of A on the imaginary axis, and by $\nu(A)$ the number of eigenvalues of A with negative real part. The **inertia** of A is defined as the triple $\text{in}(A) = (\pi(A), \nu(A), \delta(A))$.

Facts:

All the facts are proven in [OS62].

1. Let A be a complex square matrix. There exists a Hermitian matrix H such that the matrix $AH + HA^*$ is positive definite if and only if $\delta(A) = 0$. Furthermore, in such a case the inertias of A and H are the same.
2. Let $\{\lambda_1, \ldots, \lambda_n\}$ be the eigenvalues of an $n \times n$ matrix A. If $\prod_{i,j=1}^{n}(\lambda_i + \overline{\lambda_j}) \neq 0$, then for any positive definite matrix P there exists a unique Hermitian matrix H such that $AH + HA^* = P$. Furthermore, the inertias of A and H are the same.
3. Let A be a complex square matrix. We have $\delta(A) = \pi(A) = 0$ if and only if there exists an $n \times n$ positive definite Hermitian matrix such that the matrix $-(AH + HA^*)$ is positive definite.

Examples:

1. It follows from Fact 1 above that a complex square matrix A has all of its eigenvalues in the right half-plane if and only if there exists a positive definite matrix H such that the matrix $AH + HA^*$ is positive definite. This fact, associating us with the discussion of the next section, is due to Lyapunov, originally proven in [L1892] for systems of differential equations. The matrix formulation is due to [Gan60].
2. In order to demonstrate that both the existence and uniqueness claims of Fact 2 may be false without the condition on the eigenvalues, consider the matrix

$$A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix},$$

for which the condition of Fact 2 is not satisfied. One can check that the only positive definite matrices P for which the equation $AH + HA^* = P$ has Hermitian solutions are matrices of the type $P = \begin{bmatrix} p_{11} & 0 \\ 0 & p_{22} \end{bmatrix}$, $p_{11}, p_{22} > 0$. Furthermore, for $P = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}$ it is easy to verify that the Hermitian solutions of $AH + HA^* = P$ are all matrices H of the type

$$\begin{bmatrix} 1 & c \\ \bar{c} & -2 \end{bmatrix}, \quad c \in \mathbb{C}.$$

If we now choose

$$A = \begin{bmatrix} 1 & 0 \\ 0 & -2 \end{bmatrix},$$

then here the condition of Fact 2 is satisfied. Indeed, for $H = \begin{bmatrix} a & c \\ \bar{c} & b \end{bmatrix}$ we have

$$AH + HA^* = \begin{bmatrix} 2a & -c \\ -\bar{c} & -4b \end{bmatrix},$$

which can clearly be solved uniquely for any Hermitian matrix P; specifically, for $P = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}$, the unique Hermitian solution H of $AH + HA^* = P$ is $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$.

19.2 Stability

Definitions:

A complex polynomial is **negative stable [positive stable]** if its roots lie in the open left [right] half-plane.

A complex square matrix A is **negative stable [positive stable]** if its characteristic polynomial is negative stable [positive stable].

We shall use the term **stable matrix** for positive stable matrix.

For an $n \times n$ matrix A and for an integer k, $1 \leq k \leq n$, we denote by $S_k(A)$ the sum of all principal minors of A of order k.

The **Routh–Hurwitz matrix** associated with A is defined to be the matrix

$$\begin{bmatrix} S_1(A) & S_3(A) & S_5(A) & \cdot & \cdot & \cdot & \cdot & 0 & 0 \\ 1 & S_2(A) & S_4(A) & & & & & \cdot & \cdot \\ 0 & S_1(A) & S_3(A) & \cdot & & & & \cdot & \cdot \\ 0 & 1 & S_2(A) & \cdot & \cdot & & & \cdot & \cdot \\ 0 & 0 & S_1(A) & \cdot & \cdot & \cdot & & \cdot & \cdot \\ \cdot & \cdot & \cdot & & \cdot & \cdot & \cdot & 0 & 0 \\ \cdot & \cdot & \cdot & & & \cdot & \cdot & S_n(A) & 0 \\ \cdot & \cdot & \cdot & & & & \cdot & S_{n-1}(A) & 0 \\ 0 & 0 & 0 & \cdot & \cdot & \cdot & \cdot & S_{n-2}(A) & S_n(A) \end{bmatrix}.$$

A square complex matrix is a P**-matrix** if it has positive principal minors.

A square complex matrix is a P_0^+**-matrix** if it has nonnegative principal minors and at least one principal minor of each order is positive.

A principal minor of a square matrix is a **leading principal minor** if it is based on consecutive rows and columns, starting with the first row and column of the matrix.

An $n \times n$ real matrix A is **sign symmetric** if it satisfies

$$\det A[\alpha, \beta] \det A[\beta, \alpha] \geq 0, \qquad \forall \alpha, \beta \subseteq \{1, \ldots, n\},\ |\alpha| = |\beta|.$$

An $n \times n$ real matrix A is **weakly sign symmetric** if it satisfies

$$\det A[\alpha, \beta] \det A[\beta, \alpha] \geq 0, \qquad \forall \alpha, \beta \subseteq \{1, \ldots, n\},\ |\alpha| = |\beta| = |\alpha \cap \beta| + 1.$$

A square real matrix is a Z**-matrix** if it has nonpositive off-diagonal elements.

A Z-matrix with positive principal minors is an M-**matrix**. (See Section 24.5 for more information and an equivalent definition.)

Facts:

Lyapunov studied the asymptotic stability of solutions of differential systems. In 1892 he proved in his paper [L1892] a theorem which yields a necessary and sufficient condition for stability of a complex matrix. The matrix formulation of Lyapunov's Theorem is apparently due to Gantmacher [Gan60], and is given as Fact 1 below. The theorem in [Gan60] was proven for real matrices; however, as was also remarked in [Gan60], the generalization to the complex case is immediate.

1. **The Lyapunov Stability Criterion:** A complex square matrix A is stable if and only if there exists a positive definite Hermitian matrix H such that the matrix $AH + HA^*$ is positive definite.
2. [OS62] A complex square matrix A is stable if and only if for every positive definite matrix G there exists a positive definite matrix H such that the matrix $AH + HA^* = G$.
3. [R1877], [H1895] **The Routh–Hurwitz Stability Criterion:** An $n \times n$ complex matrix A with a real characteristic polynomial is stable if and only if the leading principal minors of the Routh–Hurwitz matrix associated with A are all positive.
4. [LC14] (see also [Fuj26]) **The Liénard–Chipart Stability Criterion:** Let A be an $n \times n$ complex matrix with a real characteristic polynomial. The following are equivalent:
   (a) A is stable.
   (b) $S_n(A), S_{n-2}(A), \ldots > 0$ and the odd order leading principal minors of the Routh–Hurwitz matrix associated with A are positive.
   (c) $S_n(A), S_{n-2}(A), \ldots > 0$ and the even order leading principal minors of the Routh–Hurwitz matrix associated with A are positive.
   (d) $S_n(A), S_{n-1}(A), S_{n-3}(A), \ldots > 0$ and the odd order leading principal minors of the Routh–Hurwitz matrix associated with A are positive.
   (e) $S_n(A), S_{n-1}(A), S_{n-3}(A), \ldots > 0$ and the even order leading principal minors of the Routh–Hurwitz matrix associated with A are positive.
5. [Car74] Sign symmetric P-matrices are stable.
6. [HK2003] Sign symmetric stable matrices are P-matrices.
7. [Hol99] Weakly sign symmetric P-matrices of order less than 6 are stable. Nevertheless, in general, weakly sign symetric P-matrices need not be stable.
8. (For example, [BVW78]) A Z-matrix is stable if and only if it is a P-matrix (that is, it is an M-matrix).
9. [FHR05] Let A be a stable real square matrix. Then either all the diagonal elements of A are positive or A has at least one positive diagonal element and one positive off-diagonal element.
10. [FHR05] Let ζ be an n-tuple of complex numbers, $n > 1$, consisting of real numbers and conjugate pairs. There exists a real stable $n \times n$ matrix A with exactly two positive entries such that ζ is the spectrum of A.

Examples:

1. Let

$$A = \begin{bmatrix} 2 & 2 & 3 \\ 2 & 5 & 4 \\ 3 & 4 & 5 \end{bmatrix}.$$

The Routh–Hurwitz matrix associated with A is

$$\begin{bmatrix} 12 & 1 & 0 \\ 1 & 16 & 0 \\ 0 & 12 & 1 \end{bmatrix}.$$

It is immediate to check that the latter matrix has positive leading principal minors. It, thus, follows that A is stable. Indeed, the eigenvalues of A are 1.4515, 0.0657, and 10.4828.

2. Stable matrices do not form a convex set, as is easily demonstrated by the stable matrices

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 \\ 9 & 1 \end{bmatrix},$$

whose sum $\begin{bmatrix} 2 & 1 \\ 9 & 2 \end{bmatrix}$ has eigenvalues -1 and 5. Clearly, convex sets of stable matrices do exist. An example of such a set is the set of upper (or lower) triangular matrices with diagonal elements in the open right half-plane. Nevertheless, there is no obvious link between matrix stability and convexity or conic structure. Some interesting results on stable convex hulls can be found in [Bia85], [FB87], [FB88], [CL97], and [HS90]. See also the survey in [Her98].

3. In view of Facts 5 and 7 above, it would be natural to ask whether stability of a matrix implies that the matrix is a P-matrix or a weakly sign symmetric matrix. The answer to this question is negative as is demonstrated by the matrix

$$A = \begin{bmatrix} -1 & 1 \\ -5 & 3 \end{bmatrix}.$$

The eigenvalues of A are $1 \pm i$, and so A is stable. Nevertheless, A is neither a P-matrix nor a weakly sign symmetric matrix.

4. Sign symmetric P_0^+-matrices are not necessarily stable, as is demonstrated by the sign symmetric P_0^+-matrix

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix}.$$

The matrix A is not stable, having the eigenvalues $\left\{e^{\pm\frac{2\pi i}{3}}, 1, 1, 1\right\}$.

5. A P-matrix is not necessarily stable as is demonstrated by the matrix

$$\begin{bmatrix} 1 & 0 & 3 \\ 3 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix}.$$

For extensive study of spectra of P-matrices look at [HB83], [Her83], [HJ86], [HS93], and [HK2003].

19.3 Multiplicative D-Stability

Multiplicative D-stability appears in various econometric models, for example, in the study of stability of multiple markets [Met45].

Definitions:

A real square matrix A is **multiplicative D-stable** if DA is stable for every positive diagonal matrix D.

In the literature, multiplicative D-stable matrices are usually referred to as just **D-stable** matrices.

A real square matrix A is **inertia preserving** if the inertia of AD is equal to the inertia of D for every nonsingular real diagonal matrix D.

The **graph** $G(A)$ of an $n \times n$ matrix A is the simple graph whose vertex set is $\{1, \ldots, n\}$, and where there is an edge between two vertices i and j $(i \neq j)$ if and only if $a_{ij} \neq 0$ or $a_{ji} \neq 0$. (See Chapter 28 more information on graphs.)

The matrix A is said to be **acyclic** if $G(A)$ is a forest.

Facts:

The problem of characterizing multiplicative D-stabity for certain classes and for matrices of order less than 5 is dealt with in several publications (e.g., [Cai76], [CDJ82], [Cro78], and [Joh74b]). However, in general, this problem is still open. Multiplicative D-stability is characterized in [BH84] for acyclic matrices. That result generalizes the handling of tridiagonal matrices in [CDJ82]. Characterization of multiplicative D-stability using cones is given in [HSh88]. See also the survey in [Her98].

1. Tridiagonal matrices are acyclic, since their graphs are paths or unions of disjoint paths.
2. [FF58] For a real square matrix A with positive leading principal minors there exists a positive diagonal matrix D such that DA is stable.
3. [Her92] For a complex square matrix A with positive leading principal minors there exists a positive diagonal matrix D such that DA is stable.
4. [Cro78] Multiplicative D-stable matrices are P_0^+-matrices.
5. [Cro78] A 2×2 real matrix is multiplicative D-stable if and only if it is a P_0^+-matrix.
6. [Cai76] A 3×3 real matrix A is multiplicative D-stable if and only if $A + D$ is multiplicative D-stable for every nonnegative diagonal matrix D.
7. [Joh75] A real square matrix A is multiplicative D-stable if and only if $A \pm iD$ is nonsingular for every positive diagonal matrix D.
8. (For example, [BVW78]) A Z-matrix is multiplicative D-stable if and only if it is a P-matrix (that is, it is an M-matrix).
9. [BS91] Inertia preserving matrices are multiplicative D-stable.
10. [BS91] An irreducible acyclic matrix is multiplicative D-stable if and only if it is inertia preserving.
11. [HK2003] Let A be a sign symmetric square matrix. The following are equivalent:
    (a) The matrix A is stable.
    (b) The matrix A has positive leading principal minors.
    (c) The matrix A is a P-matrix.
    (d) The matrix A is multiplicative D-stable.
    (e) There exists a positive diagonal matrix D such that the matrix DA is stable.

Examples:

1. In order to illustrate Fact 2, let

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 4 & 1 & 2 \end{bmatrix}.$$

The matrix A is not stable, having the eigenvalues 4.0606 and $-0.0303 \pm 0.4953i$. Nevertheless, since A has positive leading minors, by Fact 2 there exists a positive diagonal matrix D such that the matrix DA is stable. Indeed, the eigenvalues of

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0.1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 4 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0.4 & 0.1 & 0.2 \end{bmatrix}$$

are 1.7071, 0.2929, and 0.2.

2. In order to illustrate Fact 4, let

$$A = \begin{bmatrix} 1 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix}.$$

   The matrix A is stable, having the eigenvalues $0.3376 \pm 0.5623i$ and 2.3247. Yet, we have det $A[\{2,3\}] < 0$, and so A is not a P_0^+-matrix. Indeed, observe that the matrix

$$\begin{bmatrix} 0.1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 0.1 & 0.1 & 0 \\ -1 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$

   is not stable, having the eigenvalues $-0.1540 \pm 0.1335i$ and 2.408.
3. While stability is a spectral property, and so it is always possible to check whether a given matrix is stable or not by evaluating its eigenvalues, multiplicative D-stability cannot be characterized by the spectrum of the matrix, as is demonstrated by the following two matrices

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} -1 & 2 \\ -3 & 4 \end{bmatrix}.$$

   The matrices A and B have the same spectrum. Nevertheless, while A is multiplicative D-stable, B is not, since it is not a P_0^+-matrix. Indeed, the matrix

$$\begin{bmatrix} 5 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & 2 \\ -3 & 4 \end{bmatrix} = \begin{bmatrix} -5 & 10 \\ -3 & 4 \end{bmatrix}$$

   has eigenvalues $-0.5 \pm 3.1225i$.
4. It is shown in [BS91] that the converse of Fact 9 is not true, using the following example from [Har80]:

$$A = \begin{bmatrix} 1 & 0 & -50 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}.$$

   The matrix A is multiplicative D-stable (by the characterization of 3×3 multiplicative D-stable matrices, proven in [Cai76]). However, for $D = \text{diag}\,(-1, 3, -1)$ the matrix AD is stable and, hence, A is not inertia preserving. In fact, it is shown in [BS91] that even P-matrices that are both D-stable and Lyapunov diagonally semistable (see section 19.5) are not necessarily inertia preserving.

19.4 Additive *D*-Stability

Applications of additive D-stability may be found in linearized biological systems, e.g., [Had76].

Definitions:

A real square matrix A is said to be **additive** D**-stable** if $A + D$ is stable for every nonnegative diagonal matrix D.

In some references additive D-stable matrices are referred to as **strongly stable** matrices.

Facts:

The problem of characterizing additive D-stability for certain classes and for matrices of order less than 5 is dealt with in several publications (e.g., [Cai76], [CDJ82], [Cro78], and [Joh74b]). However, in general, this problem is still open. Additive D-stability is characterized in [Her86] for acyclic matrices. That result generalizes the handling of tridiagonal matrices in [Car84].

1. [Cro78] Additive D-stable matrices are P_0^+-matrices.
2. [Cro78] A 2×2 real matrix is additive D-stable if and only if it is a P_0^+-matrix.
3. [Cro78] A 3×3 real matrix A is additive D-stable if and only if it is a P_0^+-matrix and stable.
4. (For example, [BVW78]) A Z-matrix is additive D-stable if and only if it is a P-matrix (that is, it is an M-matrix).
5. An additive D-stable matrix need not be multiplicative D-stable (cf. Example 3).
6. [Tog80] A multiplicative D-stable matrix need not be additive D-stable.

Examples:

1. In order to illustrate Fact 1, let
$$A = \begin{bmatrix} 1 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix}.$$
The matrix A is stable, having the eigenvalues $0.3376 \pm 0.5623i$ and 2.3247. Yet, we have $\det A[2,3|2,3] < 0$, and so A is not a P_0^+-matrix. Indeed, observe that the matrix
$$\begin{bmatrix} 1 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix} + \begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$
is not stable, having the eigenvalues $2.5739 \pm 0.3690i$ and -0.1479.
2. While stability is a spectral property, and so it is always possible to check whether a given matrix is stable or not by evaluating its eigenvalues, additive D-stability cannot be characterized by the spectrum of the matrix, as is demonstrated by the following two matrices:
$$A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} -1 & 2 \\ -3 & 4 \end{bmatrix}.$$
The matrices A and B have the same spectrum. Nevertheless, while A is additive D-stable, B is not, since it is not a P_0^+-matrix. Indeed, the matrix
$$\begin{bmatrix} -1 & 2 \\ -3 & 4 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} -1 & 2 \\ -3 & 7 \end{bmatrix}$$
has eigenvalues -0.1623 and 6.1623.
3. In order to demonstrate Fact 5, consider the matrix
$$A = \begin{bmatrix} 0.25 & 1 & 0 \\ -1 & 0.5 & 1 \\ 2.1 & 1 & 2 \end{bmatrix},$$
which is a P_0^+ matrix and is stable, having the eigenvalues $0.0205709 \pm 1.23009i$ and 2.70886. Thus, A is additively D-stable by Fact 3. Nevertheless, A is not multiplicative D-stable, as the eigenvalues of
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} 0.25 & 1 & 0 \\ -1 & 0.5 & 1 \\ 2.1 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 0.25 & 1 & 0 \\ -5 & 2.5 & 5 \\ 8.4 & 4 & 8 \end{bmatrix}$$
are $-0.000126834 \pm 2.76183i$ and 10.7503.

19.5 Lyapunov Diagonal Stability

Lyapunov diagonally stable matrices play an important role in various applications, for example, predator–prey systems in ecology, e.g., [Goh76], [Goh77], and [RZ82]; dynamical systems, e.g., [Ara75]; and economic models, e.g., [Joh74a] and the references in [BBP78].

Definitions:

A real square matrix A is said to be **Lyapunov diagonally stable [semistable]** if there exists a positive diagonal matrix D such that $AD + DA^T$ is positive definite [semidefinite]. In this case, the matrix D is called a **Lyapunov scaling factor** of A.

In some references Lyapunov diagonally stable matrices are referred to as just **diagonally stable** matrices or as **Volterra–Lyapunov stable**.

An $n \times n$ matrix A is said to be an H**-matrix** if the **comparison matrix** $M(A)$ defined by

$$M(A)_{ij} = \begin{cases} |a_{ii}|, & i = j \\ -|a_{ij}|, & i \neq j \end{cases}$$

is an M-matrix.

A real square matrix A is said to be **strongly inertia preserving** if the inertia of AD is equal to the inertia of D for every (not necessarily nonsingular) real diagonal matrix D.

Facts:

The problem of characterizing Lyapunov diagonal stability is, in general, an open problem. It is solved in [BH83] for acyclic matrices. Lyapunov diagonal semistability of acyclic matrices is characterized in [Her88]. Characterization of Lyapunov diagonal stability and semistability using cones is given in [HSh88]; see also the survey in [Her98]. For a book combining theoretical results, applications, and examples, look at [KB00].

1. [BBP78], [Ple77] Lyapunov diagonally stable matrices are P-matrices.
2. [Goh76] A 2×2 real matrix is Lyapunov diagonally stable if and only if it is a P-matrix.
3. [BVW78] A real square matrix A is Lyapunov diagonally stable if and only if for every nonzero real symmetric positive semidefinite matrix H, the matrix HA has at least one positive diagonal element.
4. [QR65] Lyapunov diagonally stable matrices are multiplicative D-stable.
5. [Cro78] Lyapunov diagonally stable matrices are additive D-stable.
6. [AK72], [Tar71] A Z-matrix is Lyapunov diagonally stable if and only if it is a P-matrix (that is, it is an M-matrix).
7. [HS85a] An H-matrix A is Lyapunov diagonally stable if and only if A is nonsingular and the diagonal elements of A are nonnegative.
8. [BS91] Lyapunov diagonally stable matrices are strongly inertia preserving.
9. [BH83] Acyclic matrices are Lyapunov diagonally stable if and only if they are P-matrices.
10. [BS91] Acyclic matrices are Lyapunov diagonally stable if and only if they are strongly inertia preserving.

Examples:

1. Multiplicative D-stable and additive D-stable matrices are not necessarily diagonally stable, as is demonstrated by the matrix

$$\begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}.$$

2. Another example, given in [BH85] is the matrix

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 1 & 1 & 0 \\ 0 & 1 & a & b \\ 0 & 0 & -b & 0 \end{bmatrix}, \quad a \geq 1, b \neq 0,$$

which is not Lyapunov diagonally stable, but is multiplicative D-stable if and only if $a > 1$, and is additive D-stable whenever $a = 1$ and $b \neq 1$.

3. Stability is a spectral property, and so it is always possible to check whether a given matrix is stable or not by evaluating its eigenvalues; Lyapunov diagonal stability cannot be characterized by the spectrum of the matrix, as is demonstrated by the following two matrices:

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} -1 & 2 \\ -3 & 4 \end{bmatrix}.$$

The matrices A and B have the same spectrum. Nevertheless, while A is Lyapunov diagonal stable, B is not, since it is not a P-matrix. Indeed, for every positive diagonal matrix D, the element of $AD + DA^T$ in the $(1, 1)$ position is negative and, hence, $AD + DA^T$ cannot be positive definite.

4. Let A be a Lyapunov diagonally stable matrix and let D be a Lyapunov scaling factor of A. Using continuity arguments, it follows that every positive diagonal matrix that is close enough to D is a Lyapunov scaling factor of A. Hence, a Lyapunov scaling factor of a Lyapunov diagonally stable matrix is not unique (up to a positive scalar multiplication). The Lyapunov scaling factor is not necessarily unique even in cases of Lyapunov diagonally semistable matrices, as is demonstrated by the zero matrix and the following more interesting example. Let

$$A = \begin{bmatrix} 2 & 2 & 3 \\ 2 & 2 & 3 \\ 1 & 1 & 2 \end{bmatrix}.$$

One can check that $D = \operatorname{diag}(1, 1, d)$ is a scaling factor of A whenever $\frac{1}{9} \leq d \leq 1$. On the other hand, it is shown in [HS85b] that the identity matrix is the unique Lyapunov scaling factor of the matrix

$$\begin{bmatrix} 1 & 1 & 2 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 2 & 1 & 2 \\ 2 & 2 & 0 & 1 \end{bmatrix}.$$

Further study of Lyapunov scaling factors can be found in [HS85b], [HS85c], [SB87], [HS88], [SH88], [SB88], and [CHS92].

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Topics in Advanced Linear Algebra

20

Inverse Eigenvalue Problems

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In general, an **inverse eigenvalue problem** (IEP) consists of the construction of a matrix with prescribed structural and spectral constraints. This is a two-level problem: (1) on a theoretical level the target is to determine if the IEP is solvable, that is, to find necessary and sufficient conditions for the existence of at least one **solution matrix** (a matrix with the given constraints); and (2) on a practical level, the target is the effective construction of a solution matrix when the IEP is solvable. IEPs are classified into different types according to the specific constraints. We will consider three topics: IEPs with prescribed entries, nonnegative IEPs, and affine parameterized IEPs. Other important topics include pole assignment problems, Jacobi IEPs, inverse singular value problems, etc. For interested readers, we refer to the survey [CG02] where an account of IEPs with applications and extensive bibliography can be found.

20.1 IEPs with Prescribed Entries

The underlying question for an IEP with prescribed entries (PEIEPs) is to understand how the prescription of some entries of a matrix can have repercussions on its spectral properties. A classical result on this subject is the Schur–Horn Theorem allowing the construction of a real symmetric matrix with prescribed diagonal, prescribed eigenvalues, and subject to some restrictions (see Fact 1 below). Here we consider PEIEPs that require finding a matrix with some prescribed entries and with prescribed eigenvalues or characteristic polynomial; no structural constraints are imposed on the solution matrices.

Most of the facts of Sections 20.1 and 20.2 appear in [IC00], an excellent survey that describes finite step procedures for constructing solution matrices.

Definitions:

An **IEP with prescribed entries** (PEIEP) has the following standard formulation:

Given:

(a) A field F.
(b) n elements $\lambda_1, \ldots, \lambda_n$ of F (respectively, a monic polynomial $f \in F[x]$ of degree n).
(c) t elements $p_1, \ldots, p_t$ of F.
(d) A set $\mathcal{Q} = \{(i_1, j_1), \ldots, (i_t, j_t)\}$ of t positions of an $n \times n$ matrix.

Find: A matrix $A = [a_{ij}] \in F^{n\times n}$ with $a_{i_k j_k} = p_k$ for $1 \leq k \leq t$ and such that $\sigma(A) = \{\lambda_1, \ldots, \lambda_n\}$ (respectively, such that $p_A(x) = f$).

Facts: [IC00]

1. (Schur–Horn Theorem) Given any real numbers $\lambda_1 \geq \cdots \geq \lambda_n$ and $d_1 \geq \cdots \geq d_n$ satisfying

$$\sum_{i=1}^{k} \lambda_i \geq \sum_{i=1}^{k} d_i \quad \text{for } k = 1, \ldots, n-1 \quad \text{and} \quad \sum_{i=1}^{n} \lambda_i = \sum_{i=1}^{n} d_i,$$

there exists a real symmetric $n \times n$ matrix with diagonal $(d_1, \ldots, d_n)$ and eigenvalues $\lambda_1, \ldots, \lambda_n$; and any Hermitian matrix satisfies these conditions on its eigenvalues and diagonal entries.
2. A finite step algorithm is provided in [CL83] for the construction of a solution matrix for the Schur–Horn Theorem.
3. Consider the following classes of PEIEPs:

(1.1)	F	$\lambda_1, \ldots, \lambda_n$	$p_1, \ldots, p_{n-1}$	$\|\mathcal{Q}\| = n-1$
(1.2)	F	$f = x^n + c_1x^{n-1} + \cdots + c_n$	$p_1, \ldots, p_{n-1}$	$\|\mathcal{Q}\| = n-1$
(2.1)	F	$\lambda_1, \ldots, \lambda_n$	$p_1, \ldots, p_n$	$\|\mathcal{Q}\| = n$
(2.2)	F	$f = x^n + c_1x^{n-1} + \cdots + c_n$	$p_1, \ldots, p_n$	$\|\mathcal{Q}\| = n$
(3.1)	F	$\lambda_1, \ldots, \lambda_n$	$p_1, \ldots, p_{2n-3}$	$\|\mathcal{Q}\| = 2n-3$

- [dO73a] Each PEIEP of class (1.1) is solvable.
- [Dds74] Each PEIEP of class (1.2) is solvable except if all off-diagonal entries in one row or column are prescribed to be zero and f has no root on F.
- [dO73b] Each PEIEP of class (2.1) is solvable with the following exceptions: (1) all entries in the diagonal are prescribed and their sum is different from $\lambda_1 + \cdots + \lambda_n$; (2) all entries in one row or column are prescribed, with zero off-diagonal entries and diagonal entry different from $\lambda_1, \ldots, \lambda_n$; and (3) $n = 2$, $\mathcal{Q} = \{(1,2),(2,1)\}$, and $x^2 - (\lambda_1 + \lambda_2)x + p_1p_2 + \lambda_1\lambda_2 \in F[x]$ is irreducible over F.
- [Zab86] For $n > 4$, each PEIEP of class (2.2) is solvable with the following exceptions: (1) all entries in the diagonal are prescribed and their sum is different from-c_1; (2) all entries in a row or column are prescribed, with zero off-diagonal entries and diagonal entry which is not a root of f; and (3) all off-diagonal entries in one row or column are prescribed to be zero and f has no root on F. The case $n \leq 4$ is solved but there are more exceptions.
- [Her83] Each PEIEP of class (3.1) is solvable with the following exceptions: (1) all entries in the diagonal are prescribed and their sum is different from $\lambda_1 + \cdots + \lambda_n$; and (2) all entries in one row or column are prescribed, with zero off-diagonal entries and diagonal entry different from $\lambda_1, \ldots, \lambda_n$.
- [Her83] The result for PEIEPs of class (3.1) cannot be improved to $|\mathcal{Q}| > 2n-3$ since a lot of specific nonsolvable situations appear, and, therefore, a closed result seems to be quite inaccessible.
- A gradient flow approach is proposed in [CDS04] to explore the existence of solution matrices when the set of prescribed entries has arbitrary cardinality.

4. The important case $\mathcal{Q} = \{(i, j) : i \neq j\}$ is discussed in section 20.9.
5. Let $\{p_{ij} : 1 \leq i \leq j \leq n\}$ be a set of $\frac{n^2+n}{2}$ elements of a field F. Define the set $\{r_1, \ldots, r_s\}$ of all those integers r such that $p_{ij} = 0$ whenever $1 \leq i \leq r < j \leq n$. Assume that $0 = r_0 < r_1 < \cdots < r_s < r_{s+1} = n$ and define $\beta_t = \sum_{r_{t-1} < k \leq r_t} p_{kk}$ for $t = 1, \ldots, s+1$. The following PEIEPs have been solved:

   - [BGRS90] Let $\lambda_1, \ldots, \lambda_n$ be n elements of F. Then there exists $A = [a_{ij}] \in F^{n \times n}$ with $a_{ij} = p_{ij}$ for $1 \leq i \leq j \leq n$ and $\sigma(A) = \{\lambda_1, \ldots, \lambda_n\}$ if and only if $\{1, \ldots, n\}$ has a partition $N_1 \cup \cdots \cup N_{s+1}$ such that $|N_t| = r_t - r_{t-1}$ and $\sum_{k \in N_t} \lambda_k = \beta_t$ for each $t = 1, \ldots, s+1$.
   - [Sil93] Let $f \in F[x]$ be a monic polynomial of degree n. Then there exists $A = [a_{ij}] \in F^{n \times n}$ with $a_{ij} = p_{ij}$ for $1 \leq i \leq j \leq n$ and $p_A(x) = f$ if and only if $f = f_1 \cdots f_{s+1}$, where $f_t = x^{r_t - r_{t-1}} - \beta_t x^{r_t - r_{t-1} - 1} + \cdots \in F[x]$ for $t = 1, \ldots, s+1$.

6. [Fil69] Let $d_1, \ldots, d_n$ be elements of a field F, and let $A \in F^{n \times n}$ with $A \neq \lambda I_n$ for all $\lambda \in F$ and $\mathrm{tr}(A) = \sum_{i=1}^{n} d_i$. Then A is similar to a matrix with diagonal $(d_1, \ldots, d_n)$.

Examples:

1. [dO73b] *Given*:

   (a) A field F.

   (b) $\lambda_1, \ldots, \lambda_n \in F$.

   (c) $p_1, \ldots, p_n \in F$.

   (d) $\mathcal{Q} = \{(1,1), \ldots, (n,n)\}$.
   If $\sum_{i=1}^{n} \lambda_i = \sum_{i=1}^{n} p_i$, then $A = [a_{ij}] \in F^{n \times n}$ with

$$a_{ii} = p_i, \qquad a_{ij} = 0 \quad \text{if} \quad i \leq j-2,$$
$$a_{i,i+1} = \sum_{k=1}^{i} \lambda_k - \sum_{k=1}^{i} p_k, \qquad a_{ij} = p_j - \lambda_{j+1} \quad \text{if} \quad i > j,$$

   has diagonal $(p_1, \ldots, p_n)$ and its spectrum is $\sigma(A) = \{\lambda_1, \ldots, \lambda_n\}$.

20.2 PEIEPs of 2×2 Block Type

In the 1970s, de Oliveira posed the problem of determining all possible spectra of a 2×2 block matrix A or all possible characteristic polynomials of A or all possible invariant polynomials of A when some of the blocks are prescribed and the rest vary (*invariant polynomial* is a synonym for *invariant factor*, cf. Section 6.6).

Definitions:

Let F be a field and let A be the 2×2 block matrix

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \in F^{n \times n} \quad \text{with} \quad A_{11} \in F^{l \times l} \quad \text{and} \quad A_{22} \in F^{m \times m}.$$

Notation:

- $\deg(f)$: degree of $f \in F[x]$.
- $g|f$: polynomial g divides the polynomial f.
- $ip(B)$: invariant polynomials of the square matrix B.

Facts: [IC00]

1. [dO71] Let A_{11} and a monic polynomial $f \in F[x]$ of degree n be given. Let $ip(A_{11}) = g_1|\cdots|g_l$. Then $p_A(x) = f$ is possible except if $l > m$ and $g_1 \cdots g_{l-m}$ is not a divisor of f.
2. [dS79], [Tho79] Let A_{11} and n monic polynomials $f_1, \ldots, f_n \in F[x]$ with $f_1|\cdots|f_n$ and $\sum_{i=1}^n \deg(f_i) = n$ be given. Let $ip(A_{11}) = g_1|\cdots|g_l$. Then $ip(A) = f_1|\cdots|f_n$ is possible if and only if $f_i \mid g_i \mid f_{i+2m}$ for each $i = 1, \ldots, l$ where $f_k = 0$ for $k > n$.
3. [dO75] Let A_{12} and a monic polynomial $f \in F[x]$ of degree n be given. Then $p_A(x) = f$ is possible except if $A_{12} = 0$ and f has no divisor of degree l.
4. [Zab89], [Sil90] Let A_{12} and n monic polynomials $f_1, \ldots, f_n \in F[x]$ with $f_1|\cdots|f_n$ and $\sum_{i=1}^n \deg(f_i) = n$ be given. Let $r = \operatorname{rank}(A_{12})$ and s the number of polynomials in $f_1, \ldots, f_n$ which are different from 1. Then $ip(A) = f_1|\cdots|f_n$ is possible if and only if $r \leq n - s$ with the following exceptions:
   (a) $r = 0$ and $\prod_{i=1}^n f_i$ has no divisor of degree l.
   (b) $r \geq 1$, $l - r$ odd and $f_{n-s+1} = \cdots = f_n$ with f_n irreducible of degree 2.
   (c) $r = 1$ and $f_{n-s+1} = \cdots = f_n$ with f_n irreducible of degree $k \geq 3$ and $k|l$.
5. [Wim74] Let A_{11}, A_{12}, and a monic polynomial $f \in F[x]$ of degree n be given. Let $h_1|\cdots|h_l$ be the invariant factors of $\left[\, xI_l - A_{11} \mid -A_{12} \,\right]$. Then $p_A(x) = f$ is possible if and only if $h_1 \cdots h_l | f$.
6. All possible invariant polynomials of A are characterized in [Zab87] when A_{11} and A_{12} are given. The statement of this result contains a majorization inequality involving the *controllability indices* of the pair (A_{11}, A_{12}).
7. [Sil87b] Let A_{11}, A_{22}, and n elements $\lambda_1, \ldots, \lambda_n$ of F be given. Assume that $l \geq m$ and let $ip(A_{11}) = g_1|\cdots|g_l$. Then $\sigma(A) = \{\lambda_1, \ldots, \lambda_n\}$ is possible if and only if all the following conditions are satisfied:
   (a) $\operatorname{tr}(A_{11}) + \operatorname{tr}(A_{22}) = \lambda_1 + \cdots + \lambda_n$.
   (b) If $l > m$, then $g_1 \cdots g_{l-m} | (x - \lambda_1) \cdots (x - \lambda_n)$.
   (c) If $A_{11} = aI_l$ and $A_{22} = dI_m$, then there exists a permutation τ of $\{1, \ldots, n\}$ such that $\lambda_{\tau(2i-1)} + \lambda_{\tau(2i)} = a + d$ for $1 \leq i \leq m$ and $\lambda_{\tau(j)} = a$ for $2m + 1 \leq j \leq n$.
8. [Sil87a] Let A_{12}, A_{21}, and n elements $\lambda_1, \ldots, \lambda_n$ of F be given. Then $\sigma(A) = \{\lambda_1, \ldots, \lambda_n\}$ is possible except if, simultaneously, $l = m = 1$, $A_{12} = [\, b \,]$, $A_{21} = [\, c \,]$ and the polynomial $x^2 - (\lambda_1 + \lambda_2)x + bc + \lambda_1\lambda_2 \in F[x]$ is irreducible over F.
9. Let A_{12}, A_{21}, and a monic polynomial $f \in F[x]$ of degree n be given:
   - [Fri77] If F is algebraically closed then $p_A(x) = f$ is always possible.
   - [MS00] If $F = \mathbb{R}$ and $n \geq 3$ then $p_A(x) = f$ is possible if and only if either $\min\{\operatorname{rank}(A_{12}), \operatorname{rank}(A_{21})\} > 0$ or f has a divisor of degree l.
   - If $F = \mathbb{R}$, $A_{12} = [\, b \,]$, $A_{21} = [\, c \,]$ and $f = x^2 + c_1 x + c_2 \in \mathbb{R}[x]$ then $p_A(x) = f$ is possible if and only if $x^2 + c_1 x + c_2 + bc$ has a root in $\mathbb{R}$.
10. [Sil91] Let A_{11}, A_{12}, A_{22}, and n elements $\lambda_1, \ldots, \lambda_n$ of F be given. Let $k_1|\cdots|k_l$ be the invariant factors of $\left[\, xI_l - A_{11} \mid -A_{12} \,\right]$, $h_1|\cdots|h_m$ the invariant factors of $\begin{bmatrix} -A_{12} \\ xI_m - A_{22} \end{bmatrix}$, and $g = k_1 \cdots k_l h_1 \cdots h_m$. Then $\sigma(A) = \{\lambda_1, \ldots, \lambda_n\}$ is possible if and only if all the following conditions hold:
    (a) $\operatorname{tr}(A_{11}) + \operatorname{tr}(A_{22}) = \lambda_1 + \cdots + \lambda_n$.
    (b) $g | (x - \lambda_1) \cdots (x - \lambda_n)$.
    (c) If $A_{11}A_{12} + A_{12}A_{22} = \eta A_{12}$ for some $\eta \in F$, then there exists a permutation τ of $\{1, \ldots, n\}$ such that $\lambda_{\tau(2i-1)} + \lambda_{\tau(2i)} = \eta$ for $1 \leq i \leq t$ where $t = \operatorname{rank}(A_{12})$ and $\lambda_{\tau(2t+1)}, \ldots, \lambda_{\tau(n)}$ are the roots of g.
11. If a problem of block type is solved for prescribed characteristic polynomial then the solution for prescribed spectrum easily follows.

12. The book [GKvS95] deals with PEIEPs of block type from an operator point of view.
13. A description is given in [FS98] of all the possible characteristic polynomials of a square matrix with an arbitrary prescribed submatrix.

20.3 Nonnegative IEP (NIEP)

Nonnegative matrices appear naturally in many different mathematical areas, both pure and applied, such as numerical analysis, statistics, economics, social sciences, etc. One of the most intriguing problems in this field is the so-called **nonnegative IEP** (NIEP). Its origin goes back to A.N. Kolgomorov, who in 1938 posed the problem of determining which individual complex numbers belong to the spectrum of some $n \times n$ nonnegative matrix with its spectral radius normalized to be 1. Kolgomorov's problem was generalized in 1949 by H. R. Suleǐmanova, who posed the NIEP: To determine which n-tuples of complex numbers are spectra of $n \times n$ nonnegative matrices. For definitions and additional facts about nonnegative matrices, see Chapter 9.

Definitions:

Let Π_n denote the compact subset of $\mathbb{C}$ bounded by the regular n-sided polygon inscribed in the unit circle of $\mathbb{C}$ and with one vertex at $1 \in \mathbb{C}$.

Let Θ_n denote the subset of $\mathbb{C}$ composed of those complex numbers λ such that λ is an eigenvalue of some $n \times n$ row stochastic matrix.

A **circulant matrix** is a matrix in which every row is obtained by a single cyclic shift of the previous row.

Facts:

All the following facts appear in [Min88].

1. A complex nonzero number λ is an eigenvalue of a nonnegative $n \times n$ matrix with positive spectral radius ρ if and only if $\lambda/\rho \in \Theta_n$.
2. [DD45], [DD46] $\Theta_3 = \Pi_2 \cup \Pi_3$.
3. [Mir63] Each point in $\Pi_2 \cup \Pi_3 \cup \cdots \cup \Pi_n$ is an eigenvalue of a doubly stochastic $n \times n$ matrix.
4. [Kar51] The set Θ_n is symmetric relative to the real axis and is contained within the circle $|z| \leq 1$. It intersects $|z| = 1$ at the points $e^{\frac{2\pi i a}{b}}$ where a and b run over all integers satisfying $0 \leq a < b \leq n$. The boundary of Θ_n consists of the curvilinear arcs connecting these points in circular order. For $n \geq 4$, each arc is given by one of the following parametric equations:

$$z^q(z^p - t)^r = (1-t)^r,$$
$$(z^c - t)^d = (1-t)^d z^q,$$

where the real parameter t runs over the interval $0 \leq t \leq 1$, and c, d, p, q, r are natural numbers defined by certain rules (explicitly stated in [Min88]).

Examples:

1. [LL78] The circulant matrix

$$\frac{1}{3}\begin{bmatrix} 1+2r\cos\theta & 1-2r\cos(\frac{\pi}{3}+\theta) & 1-2r\cos(\frac{\pi}{3}-\theta) \\ 1-2r\cos(\frac{\pi}{3}-\theta) & 1+2r\cos\theta & 1-2r\cos(\frac{\pi}{3}+\theta) \\ 1-2r\cos(\frac{\pi}{3}+\theta) & 1-2r\cos(\frac{\pi}{3}-\theta) & 1+2r\cos\theta \end{bmatrix}$$

has spectrum $\{1, re^{i\theta}, re^{-i\theta}\}$, and it is doubly stochastic if and only if $re^{i\theta} \in \Pi_3$.

20.4 Spectra of Nonnegative Matrices

Definitions:

$\mathcal{N}_n \equiv \{\sigma = \{\lambda_1, \dots, \lambda_n\} \subset \mathbb{C} : \exists A \geq 0 \quad \text{with spectrum } \sigma\}$.
$\mathcal{R}_n \equiv \{\sigma = \{\lambda_1, \dots, \lambda_n\} \subset \mathbb{R} : \exists A \geq 0 \quad \text{with spectrum } \sigma\}$.
$\mathcal{S}_n \equiv \{\sigma = \{\lambda_1, \dots, \lambda_n\} \subset \mathbb{R} : \exists A \geq 0 \quad \text{symmetric with spectrum } \sigma\}$.
$\mathcal{R}_n^* \equiv \{(1, \lambda_2, \dots, \lambda_n) \in \mathbb{R}^n : \{1, \lambda_2, \dots, \lambda_n\} \in \mathcal{R}_n; 1 \geq \lambda_2 \geq \cdots \geq \lambda_n\}$.
$\mathcal{S}_n^* \equiv \{(1, \lambda_2, \dots, \lambda_n) \in \mathbb{R}^n : \{1, \lambda_2, \dots, \lambda_n\} \in \mathcal{S}_n; 1 \geq \lambda_2 \geq \cdots \geq \lambda_n\}$.
For any set $\sigma = \{\lambda_1, \dots, \lambda_n\} \subset \mathbb{C}$, let

$$\rho(\sigma) = \max_{1 \leq i \leq n} |\lambda_i| \quad \text{and} \quad s_k = \sum_{i=1}^{n} \lambda_i^k \quad \text{for each } k \in \mathbb{N}.$$

A set $S \subset \mathbb{R}^n$ is **star-shaped** from $p \in S$ if every line segment drawn from p to another point in S lies entirely in S.

Facts:

Most of the following facts appear in [ELN04].

1. [Joh81] If $\sigma = \{\lambda_1, \dots, \lambda_n\} \in \mathcal{N}_n$, then σ is the spectrum of a $n \times n$ nonnegative matrix with all row sums equal to $\rho(\sigma)$.
2. If $\sigma = \{\lambda_1, \dots, \lambda_n\} \in \mathcal{N}_n$, then the following conditions hold:
   (a) $\rho(\sigma) \in \sigma$.
   (b) $\overline{\sigma} = \sigma$.
   (c) $s_i \geq 0$ for $i \geq 1$.
   (d) [LL78], [Joh81] $s_k^m \leq n^{m-1} s_{km}$ for $k, m \geq 1$.
3. $\mathcal{N}_n$ is known for $n \leq 3$, $\mathcal{R}_n$ and $\mathcal{S}_n$ are known for $n \leq 4$:
   - $\mathcal{N}_2 = \mathcal{R}_2 = \mathcal{S}_2 = \{\sigma = \{\lambda_1, \lambda_2\} \subset \mathbb{R} : s_1 \geq 0\}$.
   - $\mathcal{R}_3 = \mathcal{S}_3 = \{\sigma = \{\lambda_1, \lambda_2, \lambda_3\} \subset \mathbb{R} : \rho(\sigma) \in \sigma;\ s_1 \geq 0\}$.
   - [LL78] $\mathcal{N}_3 = \{\sigma = \{\lambda_1, \lambda_2, \lambda_3\} \subset \mathbb{C} : \overline{\sigma} = \sigma;\ \rho(\sigma) \in \sigma;\ s_1 \geq 0;\ s_1^2 \leq 3 s_2\}$.
   - $\mathcal{R}_4 = \mathcal{S}_4 = \{\sigma = \{\lambda_1, \lambda_2, \lambda_3, \lambda_4\} \subset \mathbb{R} : \rho(\sigma) \in \sigma;\ s_1 \geq 0\}$.
4. (a) [JLL96] $\mathcal{R}_n$ and $\mathcal{S}_n$ are not always equal sets.
   (b) [ELN04] $\sigma = \{97, 71, -44, -54, -70\} \in \mathcal{R}_5$ but $\sigma \notin \mathcal{S}_5$.
   (c) [ELN04] provides symmetric matrices for all known elements of $\mathcal{S}_5$.
5. [Rea96] Let $\sigma = \{\lambda_1, \lambda_2, \lambda_3, \lambda_4\} \subset \mathbb{C}$ with $s_1 = 0$. Then $\sigma \in \mathcal{N}_4$ if and only if $s_2 \geq 0$, $s_3 \geq 0$ and $4 s_4 \geq s_2^2$. Moreover, σ is the spectrum of

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ \frac{s_2}{4} & 0 & 1 & 0 \\ \frac{s_3}{4} & 0 & 0 & 1 \\ \frac{4s_4 - s_2^2}{16} & \frac{s_3}{12} & \frac{s_2}{4} & 0 \end{bmatrix}.$$

6. [LM99] Let $\sigma = \{\lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5\} \subset \mathbb{C}$ with $s_1 = 0$. Then $\sigma \in \mathcal{N}_5$ if and only if the following conditions are satisfied:
   (a) $s_i \geq 0$ for $i = 2, 3, 4, 5$.
   (b) $4 s_4 \geq s_2^2$.
   (c) $12 s_5 - 5 s_2 s_3 + 5 s_3 \sqrt{4 s_4 - s_2^2} \geq 0$.
   The proof of the sufficient part is constructive.

7. (a) $\mathcal{R}_n^*$ and $\mathcal{S}_n^*$ are star-shaped from $(1, \ldots, 1)$.

   (b) [BM97] $\mathcal{R}_n^*$ is star-shaped from $(1, 0, \ldots, 0)$.

   (c) [KM01], [Mou03] $\mathcal{R}_n^*$ and $\mathcal{S}_n^*$ are not convex sets for $n \geq 5$.

Examples:

1. We show that $\sigma = \{5, 5, -3, -3, -3\} \notin \mathcal{N}_5$. Suppose A is a nonnegative matrix with spectrum σ. By the Perron–Frobenius Theorem, A is reducible and σ can be partitioned into two nonempty subsets, each one being the spectrum of a nonnegative matrix with Perron root equal to 5. This is not possible since one of the subsets must contain numbers with negative sum.
2. $\{6, 1, 1, -4, -4\} \in \mathcal{N}_5$ by Fact 6.

20.5 Nonzero Spectra of Nonnegative Matrices

For the definitions and additional facts about primitive matrices see Section 29.6 and Chapter 9.

Definitions:

The **Möbius function** $\mu : \mathbb{N} \mapsto \{-1, 0, 1\}$ is defined by $\mu(1) = 1$, $\mu(m) = (-1)^e$ if m is a product of e distinct primes, and $\mu(m) = 0$ otherwise.

The k^{th} **net trace** of $\sigma = \{\lambda_1, \ldots, \lambda_n\} \subset \mathbb{C}$ is $\text{tr}_k(\sigma) = \sum_{d|k} \mu(\frac{k}{d})\text{s}_d$.

The set $\sigma = \{\lambda_1, \ldots, \lambda_n\} \subset \mathbb{C}$ with $0 \notin \sigma$ is the **nonzero spectrum** of a matrix if there exists a $t \times t$ matrix, $t \geq n$, whose spectrum is $\{\lambda_1, \ldots, \lambda_n, 0, \ldots, 0\}$ with $t - n$ zeros.

The set $\sigma = \{\lambda_1, \ldots, \lambda_n\} \subset \mathbb{C}$ has a **Perron value** if $\rho(\sigma) \in \sigma$ and there exists a unique index i with $\lambda_i = \rho(\sigma)$.

Facts:

1. [BH91] Spectral Conjecture: Let $\mathbb{S}$ be a unital subring of $\mathbb{R}$. The set $\sigma = \{\lambda_1, \ldots, \lambda_n\} \subset \mathbb{C}$ with $0 \notin \sigma$ is the nonzero spectrum of some primitive matrix over $\mathbb{S}$ if and only if the following conditions hold:

   (a) σ has a Perron value.

   (b) All the coefficients of the polynomial $\prod_{i=1}^{n}(x - \lambda_i)$ lie in $\mathbb{S}$.

   (c) If $\mathbb{S} = \mathbb{Z}$, then $\text{tr}_k(\sigma) \geq 0$ for all positive integers k.

   (d) If $\mathbb{S} \neq \mathbb{Z}$, then $\text{s}_k \geq 0$ for all $k \in \mathbb{N}$ and $\text{s}_m > 0$ implies $\text{s}_{mp} > 0$ for all $m, p \in \mathbb{N}$.

2. [BH91] Subtuple Theorem: Let $\mathbb{S}$ be a unital subring of $\mathbb{R}$. Suppose that $\sigma = \{\lambda_1, \ldots, \lambda_n\} \subset \mathbb{C}$ with $0 \notin \sigma$ has $\rho(\sigma) = \lambda_1$ and satisfies conditions (a) to (d) of the spectral conjecture. If for some $j \leq n$ the set $\{\lambda_1, \ldots, \lambda_j\}$ is the nonzero spectrum of a nonnegative matrix over $\mathbb{S}$, then σ is the nonzero spectrum of a primitive matrix over $\mathbb{S}$.
3. The spectral conjecture is true for $\mathbb{S} = \mathbb{R}$ by the subtuple theorem.
4. [KOR00] The spectral conjecture is true for $\mathbb{S} = \mathbb{Z}$ and $\mathbb{S} = \mathbb{Q}$.
5. [BH91] The set $\sigma = \{\lambda_1, \ldots, \lambda_n\} \subset \mathbb{C}$ with $0 \notin \sigma$ is the nonzero spectrum of a positive matrix if and only if the following conditions hold:

   (a) σ has a Perron value.

   (b) All coefficients of $\prod_{i=1}^{n}(x - \lambda_i)$ are real.

   (c) $\text{s}_k > 0$ for all $k \in \mathbb{N}$.

Examples:

1. Let $\sigma_\epsilon = \{5, 4+\epsilon, -3, -3, -3\}$. Then:
   (a) σ_ϵ for $\epsilon < 0$ is not the nonzero spectrum of a nonnegative matrix since $s_1 < 0$.
   (b) σ_0 is the nonzero spectrum of a nonnegative matrix by Fact 2.
   (c) σ_1 is not the nonzero spectrum of a nonnegative matrix by arguing as in Example 1 of Section 20.4.
   (d) σ_ϵ for $\epsilon > 0, \epsilon \neq 1$, is the nonzero spectrum of a positive matrix by Fact 5.

20.6 Some Merging Results for Spectra of Nonnegative Matrices

Facts:

1. If $\{\lambda_1, \ldots, \lambda_n\} \in \mathcal{N}_n$ and $\{\mu_1, \ldots, \mu_m\} \in \mathcal{N}_m$, then $\{\lambda_1, \ldots, \lambda_n, \mu_1, \ldots, \mu_m\} \in \mathcal{N}_{n+m}$.
2. [Fie74] Let $\sigma = \{\lambda_1, \ldots, \lambda_n\} \in \mathcal{S}_n$ with $\rho(\sigma) = \lambda_1$ and $\tau = \{\mu_1, \ldots, \mu_m\} \in \mathcal{S}_m$ with $\rho(\tau) = \mu_1$. Then $\{\lambda_1 + \epsilon, \lambda_2, \ldots, \lambda_n, \mu_1 - \epsilon, \mu_2, \ldots, \mu_m\} \in \mathcal{S}_{n+m}$ for any $\epsilon \geq 0$ if $\lambda_1 \geq \mu_1$. The proof is constructive.
3. [Šmi04] Let A be a nonnegative matrix with spectrum $\{\lambda_1, \ldots, \lambda_n\}$ and maximal diagonal element d, and let $\tau = \{\mu_1, \ldots, \mu_m\} \in \mathcal{N}_m$ with $\rho(\tau) = \mu_1$. If $d \geq \mu_1$, then $\{\lambda_1, \ldots, \lambda_n, \mu_2, \ldots, \mu_m\} \in \mathcal{N}_{n+m-1}$. The proof is constructive.
4. Let $\sigma = \{\lambda_1, \ldots, \lambda_n\} \in \mathcal{N}_n$ with $\rho(\sigma) = \lambda_1$ and let $\epsilon \geq 0$. Then:
   (a) [Wuw97] $\{\lambda_1 + \epsilon, \lambda_2, \ldots, \lambda_n\} \in \mathcal{N}_n$.
   (b) If $\lambda_2 \in \mathbb{R}$, then not always $\{\lambda_1, \lambda_2 + \epsilon, \lambda_3, \ldots, \lambda_n\} \in \mathcal{N}_n$ (see the previous example).
   (c) [Wuw97] If $\lambda_2 \in \mathbb{R}$, then $\{\lambda_1 + \epsilon, \lambda_2 \pm \epsilon, \lambda_3, \ldots, \lambda_n\} \in \mathcal{N}_n$ (the proof is not constructive).

Examples:

1. Let $\sigma = \{\lambda_1, \ldots, \lambda_n\} \in \mathcal{N}_n$ with $\rho(\sigma) = \lambda_1$, and $\tau = \{\mu_1, \ldots, \mu_m\} \in \mathcal{N}_m$ with $\rho(\tau) = \mu_1$. By Fact 1 of section 20.4 there exists $A \geq 0$ with spectrum σ and row sums λ_1, and $B \geq 0$ with spectrum τ and row sums μ_1. [BMS04] If $\lambda_1 \geq \mu_1$ and $\epsilon \geq 0$, then the nonnegative matrix

$$\left[\begin{array}{c|c} A & \epsilon\, \mathbf{e}\mathbf{e}_1^T \\ \hline (\lambda_1 - \mu_1 + \epsilon)\, \mathbf{e}\mathbf{e}_1^T & B \end{array}\right] \geq 0$$

has row sums $\lambda_1 + \epsilon$ and spectrum $\{\lambda_1 + \epsilon, \lambda_2, \ldots, \lambda_n, \mu_1 - \epsilon, \mu_2, \ldots, \mu_m\}$.

20.7 Sufficient Conditions for Spectra of Nonnegative Matrices

Definitions:

The set $\{\lambda_1, \ldots, \lambda_{i-1}, \alpha, \beta, \lambda_{i+1}, \ldots, \lambda_n\}$ is a **negative subdivision** of $\{\lambda_1, \ldots, \lambda_n\}$ if $\alpha + \beta = \lambda_i$ with $\alpha, \beta, \lambda_i < 0$.

Facts:

Most of the following facts appear in [ELN04] and [SBM05].

1. [Sul49] Let $\sigma = \{\lambda_1, \ldots, \lambda_n\} \subset \mathbb{R}$ with $\lambda_1 \geq \cdots \geq \lambda_n$. Then $\sigma \in \mathcal{R}_n$ if

$$(\mathbf{Su}) \begin{cases} \bullet\ \lambda_1 \geq 0 \geq \lambda_2 \geq \cdots \geq \lambda_n \\ \bullet\ \lambda_1 + \cdots + \lambda_n \geq 0 \end{cases}.$$

2. [BMS04] Complex version of (Su). Let $\sigma = \{\lambda_1, \dots, \lambda_n\} \subset \mathbb{C}$ be a set that satisfies:
   (a) $\overline{\sigma} = \sigma$.
   (b) $\rho(\sigma) = \lambda_1$.
   (c) $\lambda_1 + \cdots + \lambda_n \geq 0$.
   (d) $\{\lambda_2, \dots, \lambda_n\} \subset \{z \in \mathbb{C} : \text{Re}z \leq 0, |\text{Re}z| \geq |\text{Im}z|\}$.

   Then $\sigma \in \mathcal{N}_n$ and the proof is constructive.
3. [Sou83] Let $\sigma = \{\lambda_1, \dots, \lambda_n\} \subset \mathbb{R}$ with $\lambda_1 \geq \cdots \geq \lambda_n$. Then there exists a symmetric doubly stochastic matrix D such that $\lambda_1 D$ has spectrum σ if

$$(\textbf{Sou}) \quad \left\{ \frac{1}{n}\lambda_1 + \frac{n-m-1}{n(m+1)}\lambda_2 + \sum_{k=1}^{m} \frac{\lambda_{n-2k+2}}{(k+1)k} \geq 0 \quad \text{where} \quad m = \left[\tfrac{n-1}{2}\right] \right.$$

   and the proof is constructive.
4. [Kel71] Let $\sigma = \{\lambda_1, \dots, \lambda_n\} \subset \mathbb{R}$ with $\lambda_1 \geq \cdots \geq \lambda_n$. Let r be the greatest index for which $\lambda_r \geq 0$ and let $\delta_i = \lambda_{n+2-i}$ for $2 \leq i \leq n-r+1$. Define $K = \{i : 2 \leq i \leq \min\{r, n-r+1\}$ and $\lambda_i + \delta_i < 0\}$. Then $\sigma \in \mathcal{R}_n$ if

$$(\textbf{Ke}) \quad \begin{cases} \bullet\ \lambda_1 + \sum_{i \in K,\, i<k}(\lambda_i + \delta_i) + \delta_k \geq 0 \text{ for all } k \in K \\ \bullet\ \lambda_1 + \sum_{i \in K}(\lambda_i + \delta_i) + \sum_{j=r+1}^{n-r+1} \delta_j \geq 0 \end{cases}.$$

5. [Bor95] Let $\sigma = \{\lambda_1, \dots, \lambda_n\} \subset \mathbb{R}$ with $\lambda_1 \geq \cdots \geq \lambda_n$. Then $\sigma \in \mathcal{R}_n$ if

$$(\textbf{Bo}) \quad \begin{cases} \bullet\ \exists\, \tau = \{\beta_1, \dots, \beta_d\} \subset \mathbb{R} \text{ with } d \leq n \text{ that satisfies conditions (Ke)} \\ \bullet\ \sigma \text{ is obtained from } \tau \text{ after } n-d \text{ negative subdivisions} \end{cases}.$$

6. [Sot03] Let $\sigma = \{\lambda_1, \dots, \lambda_n\} \subset \mathbb{R}$. For any partition $\sigma = \sigma^{(1)} \cup \cdots \cup \sigma^{(t)}$ where $\sigma^{(k)} = \{\lambda_1^{(k)}, \dots, \lambda_{n_k}^{(k)}\}$ with $\lambda_1^{(k)} \geq \cdots \geq \lambda_{n_k}^{(k)}$ define

$$\begin{aligned} &R_j^{(k)} = \lambda_j^{(k)} + \lambda_{n_k-j+1}^{(k)} \text{ for } 2 \leq j \leq \left[\tfrac{n_k}{2}\right] \text{ and } R_{\frac{n_k+1}{2}}^{(k)} = \lambda_{\frac{n_k+1}{2}}^{(k)} \quad \text{if } n_k \text{ odd;} \\ &T^{(k)} = \lambda_1^{(k)} + \lambda_{n_k}^{(k)} + \sum_{R_j^{(k)}<0} R_j^{(k)}. \end{aligned}$$

   Then $\sigma \in \mathcal{R}_n$ if

$$(\textbf{Sot}) \quad \begin{cases} \text{there exists a partition } \sigma = \sigma^{(1)} \cup \cdots \cup \sigma^{(t)} \text{ such that} \\ \lambda_1^{(1)} + \displaystyle\sum_{T^{(k)}<0,k=2}^{t} T^{(k)} \geq \max\left\{\lambda_1^{(1)} - T^{(1)}, \max_{2 \leq k \leq t}\{\lambda_1^{(k)}\}\right\} \end{cases}$$

   and the proof is constructive.
7. [SBM05] (Su) $\Rightarrow$ (Ke) $\Rightarrow$ (Bo) $\Rightarrow$ (Sot) and no opposite implication is true.
8. [Rad96] (Sou) and (Ke) are not comparable (see Example 2 below).
9. [Rad96] If σ satisfies (Bo), then $\sigma \in \mathcal{S}_n$.
10. [RS03] Let $\sigma = \{\lambda_1, \dots, \lambda_n\} \subset \mathbb{C}$ with $\overline{\sigma} = \sigma$ and $d = -\sum_{i=2}^{n} \lambda_i > 0$. Let $c_1, \dots, c_n$ be defined by $(x-d)\prod_{i=2}^{n}(x-\lambda_i) = x^n + \sum_{k=1}^{n} c_k x^{n-k}$. If $\lambda_1 \geq d\big(1 + \sum_{c_k>0} \frac{c_k}{d^k}\big)$, then $\sigma \in \mathcal{N}_n$. The proof is constructive.

Examples:

1. If $\sigma = \{\lambda_1, \dots, \lambda_n\} \subset \mathbb{R}$ satisfies (Su), then the companion matrix of the polynomial $\prod_{i=1}^{n}(x-\lambda_i)$ is nonnegative with spectrum σ.

2. $\{5, 3, -2, -2, -4\}$ satisfies (Ke), but not (Sou), and $\{5, 3, -2, -2, -2, -2\}$ satisfies (Sou), but not (Ke).
3. $\sigma = \{8, 4, -3, -3, -3, -3\}$ does not satisfies (Ke), but it satisfies (Bo) since σ is obtained from $\tau = \{8, 4, -6, -6\}$ after two negative subdivisions and τ satisfies (Ke).
4. $\sigma = \{9, 7, 2, 2, -5, -5, -5, -5\}$ does not satisfy (Bo), but it satisfies (Sot) with $\sigma^{(1)} = \{9, 2, -5, -5\}$ and $\sigma^{(2)} = \{7, 2, -5, -5\}$.
5. $\sigma = \{6, 1, 1, -4, -4\}$ does not satisfy (Sot), but $\sigma \in \mathcal{R}_5$ (Example 2 of section 20.4).

20.8 Affine Parameterized IEPs (PIEPs)

The set $F^{n\times n}$ of $n \times n$ matrices over the field F is naturally identified with the vector space F^{n^2}. An affine parameterized IEP requires finding within a given affine subspace of $F^{n\times n}$ a matrix with prescribed spectrum. Here we will consider that the given affine subspace is n-dimensional and $F = \mathbb{R}$ or $F = \mathbb{C}$. Especially interesting is the case where the affine subspace contains only real symmetric matrices. Some important motivating applications, including the Sturn–Liouville problem, inverse vibration problems, and nuclear spectroscopy are discussed in [FNO87].

Most of the facts of Sections 20.8, 20.9, and 20.10 appear in [Dai98].

Definitions:

An **affine parameterized IEP** (PIEP) has the following standard formulation:
Given: A field F; $n + 1$ matrices $A, A_1, \ldots, A_n \in F^{n\times n}$; and n elements $\lambda_1, \ldots, \lambda_n \in F$.
Find: $\mathbf{c} = (c_1, \ldots, c_n)^T \in F^n$ such that $\{\lambda_1, \ldots, \lambda_n\}$ is the spectrum of the matrix

$$A(\mathbf{c}) = A + c_1 A_1 + \cdots + c_n A_n.$$

In particular, a PIEP($\mathbb{C}$) is a PIEP with $F = \mathbb{C}$, a PIEP($\mathbb{R}$) is a PIEP with $F = \mathbb{R}$, and a PIEP($\mathbb{R}$S) is a PIEP with $F = \mathbb{R}$ and with all given matrices symmetric.

Facts: [Dai98]

1. [Xu92] Almost all PIEP($\mathbb{C}$) are solvable.
2. [SY86] Almost all PIEP($\mathbb{R}$) and almost all PIEP($\mathbb{R}$S) are unsolvable in the presence of multiple eigenvalues.
3. All known sufficient conditions for the solvability of a PIEP($\mathbb{R}$) or a PIEP($\mathbb{R}$S) require that the eigenvalues should be sufficiently pairwise separated. An account of necessary and of sufficient conditions can be found in [Dai98].

20.9 Relevant PIEPs Which are Solvable Everywhere

Definitions:

Additive IEP (AIEP): Given $A \in \mathbb{C}^{n\times n}$ and $\lambda_1, \ldots, \lambda_n \in \mathbb{C}$, find a diagonal matrix $D \in \mathbb{C}^{n\times n}$ such that $\sigma(A + D) = \{\lambda_1, \ldots, \lambda_n\}$.

Multiplicative IEP (MIEP): Given $B \in \mathbb{C}^{n\times n}$ and $\lambda_1, \ldots, \lambda_n \in \mathbb{C}$, find a diagonal matrix $D \in \mathbb{C}^{n\times n}$ such that $\sigma(BD) = \{\lambda_1, \ldots, \lambda_n\}$.

Toeplitz IEP (ToIEP): Given $\lambda_1, \ldots, \lambda_n \in \mathbb{R}$, find $\mathbf{c} = [c_1, \ldots, c_n]^T \in \mathbb{R}^n$ such that $\left[\, t_{ij} \,\right]_{i,j=1}^n$ with $t_{ij} = c_{|i-j|+1}$ has spectrum $\{\lambda_1, \ldots, \lambda_n\}$.

Facts: [Dai98]

1. Each AIEP is a PIEP($\mathbb{C}$) with $A_k = \mathbf{e}_k\mathbf{e}_k^T$ for $k = 1, \ldots, n$. Each AIEP is also an IEP with prescribed (off-diagonal) entries.
2. [Fri77] For any $A \in \mathbb{C}^{n\times n}$ and any $\lambda_1, \ldots, \lambda_n \in \mathbb{C}$, the corresponding AIEP is solvable, with the number of solutions not exceeding $n!$. Moreover, for almost all $\lambda_1, \ldots, \lambda_n$ there are exactly $n!$ solutions.
3. Each MIEP is a PIEP($\mathbb{C}$) with $A = 0$ and $A_k = \mathbf{v}_k\mathbf{e}_k^T$ for $k = 1, \ldots, n$, where $\mathbf{v}_1, \ldots, \mathbf{v}_n \in \mathbb{C}^n$ and $B = [\,\mathbf{v}_1 \ \cdots \ \mathbf{v}_n\,] \in \mathbb{C}^{n\times n}$.
4. [Fri75] Assume that all principal minors of $B \in \mathbb{C}^{n\times n}$ are nonzero. For any $\lambda_1, \ldots, \lambda_n \in \mathbb{C}$, the corresponding MIEP is solvable, with the number of solutions not exceeding $n!$. Moreover, for almost all $\lambda_1, \ldots, \lambda_n$ there are exactly $n!$ solutions.
5. Each ToIEP is a PIEP($\mathbb{R}$S) with $A = 0$ and $A_k = [\,a_{ij}^{(k)}\,]_{i,j=1}^n$, where $a_{ij}^{(k)} = 1$ if $|i - j| + 1 = k$ and $a_{ij}^{(k)} = 0$ otherwise.
6. [Lan94] For any $\lambda_1, \ldots, \lambda_n \in \mathbb{R}$ the corresponding ToIEP is solvable.

20.10 Numerical Methods for PIEPs

Facts: [Dai98]

1. For a given PIEP($\mathbb{R}$S), it is possible to order both the eigenvalues $\lambda_1 \leq \cdots \leq \lambda_n$ and the eigenvalues $\lambda_1(\mathbf{c}) \leq \cdots \leq \lambda_n(\mathbf{c})$ of $A(\mathbf{c})$. Then solving the PIEP($\mathbb{R}$S) is equivalent to solving the nonlinear system

$$f(\mathbf{c}) = (\lambda_1(\mathbf{c}) - \lambda_1, \ldots, \lambda_n(\mathbf{c}) - \lambda_n)^T = 0.$$

   Assume that a solution $\mathbf{c}^*$ exists and that the given eigenvalues are distinct:

   - **Method 1a.** Newton's method provides a locally quadratically convergent (l.q.c.) algorithm, and it is usually l.q.c. even in the presence of multiple eigenvalues ([FNO87]). Each iteration in Newton's method involves the solution of an eigenvalue–eigenvector problem.
   - **Method 1b.** A Newton-like method is given in [FNO87] . Newton's method is modified by using the inverse power method to find approximate eigenvectors in each iteration. The new algorithm maintains l.q.c. (see [CXZ99]).
   - **Method 1c.** An inexact Newton-like method is given in [CCX03] . The last iterations of the inverse power method are truncated avoiding oversolving. The algorithm converges superlinearly, but the overall cost is reduced. In particular, for ToIEPs, this improved algorithm has better performance than specific known algorithms.

2. For a given PIEP($\mathbb{R}$) or PIEP($\mathbb{C}$), complex eigenvalues can appear. Assume that for the corresponding PIEP a solution $\mathbf{c}^*$ exists:

   - **Method 2.** In [BK81], Newton's method is applied to solve

$$f(\mathbf{c}) = (\det(A(\mathbf{c}) - \lambda_1 I_n), \ldots, \det(A(\mathbf{c}) - \lambda_n I_n))^T = 0.$$

     The algorithm is l.q.c. and is suitable for the case of distinct eigenvalues.

   - **Method 3.** Newton's method is applied in [Xu96] to solve

$$f(\mathbf{c}) = (\sigma_{min}(A(\mathbf{c}) - \lambda_1 I_n), \ldots, \sigma_{min}(A(\mathbf{c}) - \lambda_n I_n))^T = 0,$$

     where σ_{min} denotes the smallest singular value. The algorithm is l.q.c. under mild conditions even when multiple eigenvalues are present.

- **Method 4a.** An l.q.c. algorithm based on the QR decomposition theory is given in [Li92] that is suitable for the case of distinct eigenvalues.
- **Method 4b.** Method 4a is extended in [Dai99] to the case of multiple eigenvalues for any PIEP($\mathbb{R}$S). The new algorithm is l.q.c. and is based on QR-like decomposition theory and least square techniques. Methods 4a and 4b are given in a more general context.

3. All previous methods require starting from an initial point close to a solution in order to guarantee convergence.

   - **Method 5.** A homotopy approach has been considered for complex symmetric matrices (see [Chu90], [Xu93]), which in theory provides a globally convergent algorithm by which all solutions can be found.

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21

Totally Positive and Totally Nonnegative Matrices

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Total positivity has been a recurring theme in linear algebra and other aspects of mathematics for the past 80 years. Totally positive matrices, in fact, originated from studying small oscillations in mechanical systems [GK60], and from investigating relationships between the number of sign changes of the vectors $\mathbf{x}$ and $A\mathbf{x}$ for fixed A [Sch30]. Since then this class (and the related class of sign-regular matrices) has arisen in such a wide range of applications (see [GM96] for an incredible compilation of interesting articles dealing with a long list of relevant applications of totally positive matrices) that over the years many convenient points of view for total positivity have been offered and later defended by many prominent mathematicians.

After F.R. Gantmacher and M.G. Krein [GK60], totally positive matrices were seen and further developed in connection with spline functions, collocation matrices, and generating totally positive sequences (Polya frequency sequences). Then in 1968 came one of the most important and influential references in this area, namely the book *Total Positivity* by S. Karlin [Kar68]. Karlin approached total positivity by considering the analytic properties of totally positive functions. Along these lines, he studied totally positive kernels, sign-regular functions, and Polya frequency functions. Karlin also notes the importance of total positivity in the field of statistics.

The next significant view point can be seen in T. Ando's survey paper [And87]. Ando's contribution was to consider a multilinear approach to this subject, namely making use of skew-symmetric products and Schur complements as his underlying tools. More recently, it has become clear that factorizations of totally positive matrices are a fruitful avenue for research on this class. Coupled with matrix factorizations, totally positive matrices have taken on a new combinatorial form known as Planar networks. Karlin and G. McGregor [KM59] were some of the pioneers of this view point on total positivity (see also [Bre95][Lin73]), and there has since been a revolution of sorts with many new and exciting advances and additional applications (e.g., positive elements in reductive Lie groups, computer-aided geometric design, shape-preserving designs).

This area is not only a historically significant one in linear algebra, but it will continue to produce many important advances and spawn many more worthwhile applications.

21.1 Basic Properties

In this section, we present many basic, yet fundamental, properties associated with the important class of totally positive and totally nonnegative matrices.

Definitions:

An $m \times n$ real matrix A is **totally nonnegative** (TN) if the determinant of every square submatrix (i.e., minor) is nonnegative.

An $m \times n$ real matrix A is **totally positive** (TP) if every minor of A is positive.

An $n \times n$ matrix A is **oscillatory** if A is totally nonnegative and A^k is totally positive for some integer $k \geq 1$.

An $m \times n$ real matrix is **in double echelon form** if

(a) Each row of A has one of the following forms ($*$ indicates a nonzero entry):

1. $(*, *, \cdots, *)$,
2. $(*, \cdots, *, 0, \cdots, 0)$,
3. $(0, \cdots, 0, *, \cdots, *)$, or
4. $(0, \cdots, 0, *, \cdots, *, 0, \cdots, 0)$.

(b) The first and last nonzero entries in row $i + 1$ are not to the left of the first and last nonzero entries in row i, respectively ($i = 1, 2, \dots, n - 1$).

Facts:

1. Every totally positive matrix is a totally nonnegative matrix.
2. Suppose A is a totally nonnegative (positive) rectangular matrix. Then
   (a) A^T, the transpose of A, is totally nonnegative (positive),
   (b) $A[\alpha, \beta]$ is totally nonnegative (positive) for any row index set α and column index set β.
3. (Section 4.2) Cauchy–Binet Identity: Since any $k \times k$ minor of the product AB is a sum of products of $k \times k$ minors of A and B, it follows that if all the $k \times k$ minors of two $n \times n$ matrices are positive, then all the $k \times k$ minors of their product are positive.
4. [GK02, p .74], [And87] The set of all totally nonnegative (positive) matrices is closed under multiplication.
5. Let A be TP (TN) and D_1 and D_2 be positive diagonal matrices. Then $D_1 A D_2$ is TP (TN).
6. [GK02, p .75] If A is a square invertible totally nonnegative (or is totally positive) matrix, then $SA^{-1}S$ is totally nonnegative (positive) for $S = \text{diag}(1, -1, \cdots, \pm 1)$. Hence, if A is a square invertible totally nonnegative matrix (or is totally positive), then the unsigned adjugate matrix of A (or the $(n-1)$st compound of A) is totally nonnegative (positive).
7. [And87], [Fal99] If A is a square totally nonnegative (positive) matrix, then, assuming $A[\alpha^c]$ is invertible, $A/A[\alpha^c] = A[\alpha] - A[\alpha, \alpha^c](A[\alpha^c])^{-1}A[\alpha^c, \alpha]$, the Schur complement of $A[\alpha^c]$ in A, is totally nonnegative (positive), for all index sets α based on consecutive indices. Recall that α^c denotes the complement of the set α.
8. [And87], [Fal01], [Whi52], [Kar68, p. 98] The closure of the totally positive matrices (in the usual topology on $\mathbb{R}^{m \times n}$) is the totally nonnegative matrices.
9. [Fal99], [JS00] Let $A = [\mathbf{a_1}, \mathbf{a_2}, \dots, \mathbf{a_n}]$ be an $m \times n$ totally nonnegative (positive) matrix whose i^{th} column is $\mathbf{a_i}$ ($i = 1, 2, \dots, n$). Suppose C denotes the set of all column vectors $\mathbf{b}$ for which

the $m \times (n+1)$ matrix $\hat{A} = [\mathbf{a_1}, \ldots, \mathbf{a_k}, \mathbf{b}, \mathbf{a_{k+1}}, \ldots, \mathbf{a_n}]$ is a totally nonnegative (positive) matrix (here k is fixed but arbitrary). Then C is a nonempty convex cone.

10. [Fal99] If A is an $m \times n$ totally nonnegative (positive) matrix, then increasing the (1,1) or the (m, n) entries of A results in a totally nonnegative (positive) matrix. In general these are the only two entries in a TN matrix with this property (see [FJS00]).
11. [And87] Let P denote the $n \times n$ permutation matrix induced by the permutation $i \to n - i + 1$, $(1 \leq i \leq n)$, and suppose A is an $n \times n$ totally nonnegative (positive) matrix. Then PAP is a totally nonnegative (positive) matrix.
12. Any irreducible tridiagonal matrix with nonzero main diagonal is in double echelon form.
13. [Fal99] Let A be an $m \times n$ totally nonnegative matrix with no zero rows or columns. Then A is in double echelon form.
14. [Rad68], [Fal99] An $n \times n$ totally nonnegative matrix $A = [a_{ij}]$ is irreducible if and only if $a_{i,i+1} > 0$ and $a_{i+1,i} > 0$, for $i = 1, 2, \ldots, n-1$.

Examples:

1. Consider the following 3×3 matrix: $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{bmatrix}$. It is not difficult to check that all minors of A are positive.
2. (Inverse tridiagonal matrix) From Fact 6 above, the inverse of a TN tridiagonal matrix is signature similar to a TN matrix. Such matrices are referred to as "single-pair" matrices in [GK02, pp. 78–80], are very much related to "Green's matrices" (see [Kar68, pp. 110–112]), and are similar to matrices of type D found in [Mar70a].
3. (Vandermonde matrix) Vandermonde matrices arise in the problem of determining a polynomial of degree at most $n-1$ that interpolates n data points. Suppose that n data points $(x_i, y_i)_{i=1}^n$ are given. The goal is to construct a polynomial $p(x) = a_0 + a_1 x + \cdots + a_{n-1} x^{n-1}$ that satisfies $p(x_i) = y_i$ for $i = 1, 2, \ldots, n$, which can be expressed as

$$\begin{bmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \cdots & x_2^{n-1} \\ \vdots & & \vdots & & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^{n-1} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_{n-1} \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}. \tag{21.1}$$

   The $n \times n$ coefficient matrix in (21.1) is called a *Vandermonde matrix*, and we denote it by $\mathcal{V}(x_1, \ldots, x_n)$. The determinant of the $n \times n$ Vandermonde matrix in (21.1) is given by the formula $\prod_{i>j}(x_i - x_j)$; see [MM64, pp. 15–16]. Thus, if $0 < x_1 < x_2 < \cdots < x_n$, then $\mathcal{V}(x_1, \ldots, x_n)$ has positive entries, positive leading principal minors, and positive determinant. More generally, it is known [GK02, p. 111] that if $0 < x_1 < x_2 < \cdots < x_n$, then $\mathcal{V}(x_1, \ldots, x_n)$ is TP. Example 1 above is a Vandermonde matrix.
4. Let $f(x) = \sum_{i=0}^n a_i x^i$ be an n^{th} degree polynomial in x. The *Routh–Hurwitz matrix* is the $n \times n$ matrix given by

$$A = \begin{bmatrix} a_1 & a_3 & a_5 & a_7 & \cdots & 0 & 0 \\ a_0 & a_2 & a_4 & a_6 & \cdots & 0 & 0 \\ 0 & a_1 & a_3 & a_5 & \cdots & 0 & 0 \\ 0 & a_0 & a_2 & a_4 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \cdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & a_{n-1} & 0 \\ 0 & 0 & 0 & 0 & \cdots & a_{n-2} & a_n \end{bmatrix}.$$

A specific example of a Routh–Hurwitz matrix for an arbitrary polynomial of degree six, $f(x) = \sum_{i=0}^{6} a_i x^i$, is given by

$$A = \begin{bmatrix} a_1 & a_3 & a_5 & 0 & 0 & 0 \\ a_0 & a_2 & a_4 & a_6 & 0 & 0 \\ 0 & a_1 & a_3 & a_5 & 0 & 0 \\ 0 & a_0 & a_2 & a_4 & a_6 & 0 \\ 0 & 0 & a_1 & a_3 & a_5 & 0 \\ 0 & 0 & a_0 & a_2 & a_4 & a_6 \end{bmatrix}.$$

A polynomial $f(x)$ is *stable* if all the zeros of $f(x)$ have negative real parts. It is proved in [Asn70] that $f(x)$ is stable if and only if the Routh–Hurwitz matrix formed from f is totally nonnegative.

5. (Cauchy matrix) An $n \times n$ matrix $C = [c_{ij}]$ is called a *Cauchy matrix* if the entries of C are given by

$$c_{ij} = \frac{1}{x_i + y_j},$$

where $x_1, x_2, \ldots, x_n$ and $y_1, y_2, \ldots, y_n$ are two sequences of numbers (chosen so that c_{ij} is well-defined). A Cauchy matrix is totally positive if and only if $0 < x_1 < x_2 < \cdots < x_n$ and $0 < y_1 < y_2 < \cdots < y_n$ ([GK02, pp. 77–78]).

6. (Pascal matrix) Consider the 4×4 matrix $P_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 1 & 3 & 6 & 10 \\ 1 & 4 & 10 & 20 \end{bmatrix}$. The matrix P_4 is called the *symmetric* 4×4 *Pascal matrix* because of its connection with Pascal's triangle (see Example 4 of section 21.2 for a definition of P_n for general n). Then P_4 is TP, and the inverse of P_4 is given by

$$P_4^{-1} = \begin{bmatrix} 4 & -6 & 4 & -1 \\ -6 & 14 & -11 & 3 \\ 4 & -11 & 10 & -3 \\ -1 & 3 & -3 & 1 \end{bmatrix}.$$

Notice that the inverse of the 4×4 Pascal matrix is integral. Moreover, deleting the signs by forming $SP_4^{-1}S$, where $S = \text{diag}(1, -1, 1, -1)$, results in the TP matrix $\begin{bmatrix} 4 & 6 & 4 & 1 \\ 6 & 14 & 11 & 3 \\ 4 & 11 & 10 & 3 \\ 1 & 3 & 3 & 1 \end{bmatrix}$.

Applications:

1. (Tridiagonal matrices) When Gantmacher and Krein were studying the oscillatory properties of an elastic segmental continuum (no supports between the endpoints a and b) under small transverse oscillations, they were able to generate a system of linear equations that define the frequency of the oscillation (see [GK60]). The system of equations thus found can be represented in what is known as the influence-coefficient matrix, whose properties are analogous to those governing the segmental continuum. This process of obtaining the properties of the segmental continuum from the influence-coefficient matrix was only possible due to the inception of the theory of oscillatory matrices. A special case involves tridiagonal matrices (or Jacobi matrices as they were called in [GK02]). Tridiagonal matrices are not only interesting in their own right as a model example of oscillatory matrices, but they also naturally arise in studying small oscillations in certain mechanical systems, such as torsional oscillations of a system of disks fastened to a shaft. In [GK02, pp. 81–82] they prove that an irreducible tridiagonal matrix is totally nonnegative if and only if its entries are nonnegative and its leading principal minors are nonnegative.

21.2 Factorizations

Recently, there has been renewed interest in total positivity partly motivated by the so-called "bidiagonal factorization," namely, the fact that any totally positive matrix can be factored into entry-wise nonnegative bidiagonal matrices. This result has proven to be a very useful and tremendously powerful property for this class. (See Section 1.6 for basic information on LU factorizations.)

Definitions:

An **elementary bidiagonal matrix** is an $n \times n$ matrix whose main diagonal entries are all equal to one, and there is, at most, one nonzero off-diagonal entry and this entry must occur on the super- or subdiagonal. The lower elementary bidiagonal matrix whose elements are given by

$$c_{ij} = \begin{cases} 1, & \text{if } i = j, \\ \mu, & \text{if } i = k, \; j = k-1, \\ 0, & \text{otherwise} \end{cases}$$

is denoted by $E_k(\mu) = [c_{ij}]$ $(2 \leq k \leq n)$.

A triangular matrix is **ΔTP** if all of its nontrivial minors are positive. (Here a trivial minor is one which is zero only because of the zero pattern of a triangular matrix.)

Facts:

1. $(E_k(\mu))^{-1} = E_k(-\mu)$.
2. [Cry73] Let A be an $n \times n$ matrix. Then A is totally positive if and only if A has an LU factorization such that both L and U are $n \times n$ ΔTP matrices.
3. [And87], [Cry76] Let A be an $n \times n$ matrix. Then A is totally nonnegative if and only if A has an LU factorization such that both L and U are $n \times n$ totally nonnegative matrices.
4. [Whi52] Suppose $A = [a_{ij}]$ is an $n \times n$ matrix with $a_{j1}, a_{j+1,1} > 0$, and $a_{k1} = 0$ for $k > j+1$. Let B be the $n \times n$ matrix obtained from A by using row j to eliminate $a_{j+1,1}$. Then A is TN if and only if B is TN. Note that B is equal to $E_{j+1}(-a_{j+1,1}/a_{j1})A_j$ and, hence, $A = (E_{j+1}(-a_{j+1,1}/a_{j1}))^{-1}B = E_{j+1}(a_{j+1,1}/a_{j1})B$.
5. [Loe55], [GP96], [BFZ96], [FZ00], [Fal01] Let A be an $n \times n$ nonsingular totally nonnegative matrix. Then A can be written as

$$\begin{aligned} A = {} & (E_2(l_k))(E_3(l_{k-1})E_2(l_{k-2})) \cdots (E_n(l_{n-1}) \cdots E_3(l_2)E_2(l_1))D \\ & (E_2^T(u_1)E_3^T(u_2) \cdots E_n^T(u_{n-1})) \cdots (E_2^T(u_{k-2})E_3^T(u_{k-1}))(E_2^T(u_k)), \end{aligned} \tag{21.2}$$

   where $k = \binom{n}{2}$; $l_i, u_j \geq 0$ for all $i, j \in \{1, 2, \ldots, k\}$; and D is a positive diagonal matrix.
6. [Cry76] Any $n \times n$ totally nonnegative matrix A can be written as

$$A = \prod_{i=1}^{M} L^{(i)} \prod_{j=1}^{N} U^{(j)}, \tag{21.3}$$

   where the matrices $L^{(i)}$ and $U^{(j)}$ are, respectively, lower and upper bidiagonal totally nonnegative matrices with at most one nonzero entry off the main diagonal.
7. [Cry76], [RH72] If A is an $n \times n$ totally nonnegative matrix, then there exists a totally nonnegative matrix S and a tridiagonal totally nonnegative matrix T such that

   (a) $TS = SA$.

   (b) The matrices A and T have the same eigenvalues.

   Moreover, if A is nonsingular, then S is nonsingular.

Examples:

1. Let P_4 be the matrix given in Example 6 of section 21.1. Then P_4 is TP, and a (unique up to a positive diagonal scaling) LU factorization of P_4 is given by

$$P_4 = LU = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 1 & 3 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Observe that the rows of L, or the columns of U, come from the rows of Pascal's triangle (ignoring the zeros); hence, the name Pascal matrix (see Example 4 for a definition of P_n).

2. The 3×3 Vandermonde matrix A in Example 1 of Section 21.1 can be factored as

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}. \tag{21.4}$$

3. In fact, we can write $\mathcal{V}(x_1, x_2, x_3)$ as

$$\mathcal{V}(x_1,x_2,x_3) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & \frac{(x_3-x_2)}{(x_2-x_1)} & 1 \end{bmatrix}$$
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & x_2 - x_1 & 0 \\ 0 & 0 & (x_3-x_2)(x_3-x_1) \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & x_2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & x_1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & x_1 \\ 0 & 0 & 1 \end{bmatrix}.$$

4. Consider the factorization (21.2) from Fact 5 of a 4×4 matrix in which all of the variables are equal to one. The resulting matrix is P_4, which is necessarily TP. On the other hand, consider the $n \times n$ matrix $P_n = [p_{ij}]$ whose first row and column entries are all ones, and for $2 \leq i, j \leq n$ let $p_{ij} = p_{i-1,j} + p_{i,j-1}$. In fact, the relation $p_{ij} = p_{i-1,j} + p_{i,j-1}$ implies [Fal01] that P_n can be written as

$$P_n = E_n(1) \cdots E_2(1) \begin{bmatrix} 1 & 0 \\ 0 & P_{n-1} \end{bmatrix} E_2^T(1) \cdots E_n^T(1).$$

Hence, by induction, P_n has the factorization (21.2) in which the variables involved are all equal to one. Consequently, the symmetric Pascal matrix P_n is TP for all $n \geq 1$. Furthermore, since in general $(E_k(\mu))^{-1} = E_k(-\mu)$ (Fact 1), it follows that P_n^{-1} is not only signature similar to a TP matrix, but it is also integral.

21.3 Recognition and Testing

In practice, how can one determine if a given $n \times n$ matrix is TN or TP? One could calculate *every* minor, but that would involve evaluating $\sum_{k=1}^{n} \binom{n}{k}^2 \sim 4^n/\sqrt{\pi n}$ determinants. Is there a smaller collection of minors whose nonnegativity or positivity implies the nonnegativity or positivity of all minors?

Definitions:

For $\alpha = \{i_1, i_2, \dots, i_k\} \subseteq N = \{1, 2, \dots, n\}$, with $i_1 < i_2 < \cdots < i_k$, the **dispersion of** α, denoted by $d(\alpha)$, is defined to be $\sum_{j=1}^{k-1}(i_{j+1} - i_j - 1) = i_k - i_1 - (k-1)$, with the convention that $d(\alpha) = 0$, when α is a singleton.

If α and β are two contiguous index sets with $|\alpha| = |\beta| = k$, then the minor $\det A[\alpha, \beta]$ is called **initial** if α or β is $\{1, 2, \dots, k\}$. A minor is called a **leading principal minor** if it is an initial minor with both $\alpha = \beta = \{1, 2, \dots, k\}$.

An upper right (lower left) **corner minor** of A is one of the form $\det A[\alpha, \beta]$ in which α consists of the first k (last k) and β consists of the last k (first k) indices, $k = 1, 2, \dots, n$.

Facts:

1. The dispersion of a set α represents a measure of the "gaps" in the set α. In particular, observe that $d(\alpha) = 0$ if and only if α is a contiguous subset of N.
2. [Fek13] (Fekete's Criterion) An $m \times n$ matrix A is totally positive if and only if $\det A[\alpha, \beta] > 0$, for all $\alpha \subseteq \{1, 2, \dots, m\}$ and $\beta \subseteq \{1, 2, \dots, n\}$, with $|\alpha| = |\beta|$ and $d(\alpha) = d(\beta) = 0$. (Reduces the number of minors to be checked for total positivity to roughly n^3.)
3. [GP96], [FZ00] If all initial minors of A are positive, then A is TP. (Reduces the number of minors to be checked for total positivity to n^2.)
4. [SS95], [Fal04] Suppose that A is TN. Then A is TP if and only if all corner minors of A are positive.
5. [GP96] Let $A \in \mathbb{R}^{n \times n}$ be nonsingular.
   (a) A is TN if and only if for each $k = 1, 2, \dots, n$,
      i. $\det A[\{1, 2, \dots, k\}] > 0$.
      ii. $\det A[\alpha, \{1, 2, \dots, k\}] \geq 0$, for every $\alpha \subseteq \{1, 2, \dots, n\}$, $|\alpha| = k$.
      iii. $\det A[\{1, 2, \dots, k\}, \beta] \geq 0$, for every $\beta \subseteq \{1, 2, \dots, n\}$, $|\beta| = k$.
   (b) A is TP if and only if for each $k = 1, 2 \dots, n$,
      i. $\det A[\alpha, \{1, 2, \dots, k\}] > 0$, for every $\alpha \subseteq \{1, 2, \dots, n\}$ with $|\alpha| = k$, $d(\alpha) = 0$.
      ii. $\det A[\{1, 2, \dots, k\}, \beta] > 0$, for every $\beta \subseteq \{1, 2, \dots, n\}$ with $|\beta| = k$, $d(\beta) = 0$.
6. [GK02, p. 100] An $n \times n$ totally nonnegative matrix $A = [a_{ij}]$ is oscillatory if and only if
   (a) A is nonsingular.
   (b) $a_{i,i+1} > 0$ and $a_{i+1,i} > 0$, for $i = 1, 2, \dots, n-1$.
7. [Fal04] Suppose A is an $n \times n$ invertible totally nonnegative matrix. Then A is oscillatory if and only if a parameter from at least one of the bidiagonal factors E_k and E_k^T is positive, for each $k = 2, 3, \dots, n$ in the elementary bidiagonal factorization of A given in Fact 5 of section 21.2.

Examples:

1. Unfortunately, Fekete's Criterion, Fact 2, does not hold in general if "totally positive" is replaced with "totally nonnegative" and "> 0" is replaced with "≥ 0." Consider the following simple example: $A = \begin{bmatrix} 1 & 0 & 2 \\ 1 & 0 & 1 \\ 2 & 0 & 1 \end{bmatrix}$. It is not difficult to verify that every minor of A based on contiguous row and column sets is nonnegative, but $\det A[\{1, 3\}] = -3$. For an invertible and irreducible example consider $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$.

21.4 Spectral Properties

Approximately 60 years ago, Gantmacher and Krein [GK60], who were originally interested in oscillation dynamics, undertook a careful study into the theory of totally nonnegative matrices. Of the many topics they considered, one was the properties of the eigenvalues of totally nonnegative matrices.

Facts:

1. [GK02, pp. 86–91] Let A be an $n \times n$ oscillatory matrix. Then the eigenvalues of A are positive, real, and distinct. Moreover, an eigenvector $\mathbf{x_k}$ corresponding to the k^{th} largest eigenvalue has exactly $k-1$ variations in sign for $k = 1, 2, \ldots, n$. Furthermore, assuming we choose the first entry of each eigenvector to be positive, the positions of the sign change in each successive eigenvector interlace. (See Preliminaries for the definition of interlace.)
2. [And87] Let A be an $n \times n$ totally nonnegative matrix. Then the eigenvalues of A are real and nonnegative.
3. [FGJ00] Let A be an $n \times n$ irreducible totally nonnegative matrix. Then the positive eigenvalues of A are distinct.
4. [GK02, pp. 107–108] If A is an $n \times n$ oscillatory matrix, then the eigenvalues of A are distinct and strictly interlace the eigenvalues of the two principal submatrices of order $n-1$ obtained from A by deleting the first row and column or the last row and column. If A is an $n \times n$ TN matrix, then nonstrict interlacing holds between the eigenvalues of A and the two principal submatrices of order $n-1$ obtained from A by deleting the first row and column or the last row and column.
5. [Pin98] If A is an $n \times n$ totally positive matrix with eigenvalues $\lambda_1 > \lambda_2 > \cdots > \lambda_n$ and $A(k)$ is the $(n-1) \times (n-1)$ principal submatrix obtained from A by deleting the kth row and column with eigenvalues $\mu_1 > \mu_2 > \cdots > \mu_{n-1}$, then for $j = 1, 2, \ldots, n-1$, $\lambda_{j-1} > \mu_j > \lambda_{j+1}$, where $\lambda_0 = \lambda_1$. In the usual Cauchy interlacing inequalities [MM64, p. 119] for positive semidefinite matrices, λ_{j-1} is replaced by λ_j. The nonstrict inequalities need not hold for TN matrices. The extreme cases $(j = 1, n-1)$ of this interlacing result were previously proved in [Fri85].
6. [Gar82] Let $n \geq 2$ and $A = [a_{ij}]$ be an oscillatory matrix. Then the main diagonal entries of A are majorized by the eigenvalues of A. (See Preliminaries for the definition of majorization.)

Examples:

1. Consider the 4×4 TP matrix $P_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 1 & 3 & 6 & 10 \\ 1 & 4 & 10 & 20 \end{bmatrix}$. Then the eigenvalues of P_4 are 26.305, 2.203, .454, and .038, with respective eigenvectors
$$\begin{bmatrix} .06 \\ .201 \\ .458 \\ .864 \end{bmatrix}, \begin{bmatrix} .53 \\ .64 \\ .392 \\ -.394 \end{bmatrix}, \begin{bmatrix} .787 \\ -.163 \\ -.532 \\ .265 \end{bmatrix}, \begin{bmatrix} .309 \\ -.723 \\ .595 \\ -.168 \end{bmatrix}.$$
2. The irreducible, singular TN (Hessenberg) matrix given by $H = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$ has eigenvalues equal to 3, 1, 0, 0. Notice the positive eigenvalues of H are distinct.
3. Using the TP matrix P_4 in Example 1, the eigenvalues of $P_4(\{1\})$ are 26.213, 1.697, and .09. Observe that the usual Cauchy interlacing inequalities are satisfied in this case.

21.5 Deeper Properties

In this section, we explore more advanced topics that are not only interesting in their own right, but continue to demonstrate the delicate structure of these matrices.

Definitions:

For a given vector $\mathbf{c} = (c_1, c_2, \ldots, c_n)^T \in \mathbb{R}^n$ we define two quantities associated with the number of sign changes of the vector $\mathbf{c}$. These are:

$V^-(\mathbf{c})$ — the **number of sign changes in the sequence** $c_1, c_2, \ldots, c_n$ with the zero elements discarded; and

$V^+(\mathbf{c})$ — the **maximum number of sign changes in the sequence** $c_1, c_2, \ldots, c_n$, where the zero elements are arbitrarily assigned the values $+1$ and -1.

For example, $V^-((1, 0, 1, -1, 0, 1)^T) = 2$ and $V^+((1, 0, 1, -1, 0, 1)^T) = 4$.

We use $<, \leq$ to denote the usual entry-wise partial order on matrices; i.e., for $A = [a_{ij}]$, $B = [b_{ij}] \in \mathbb{R}^{m\times n}$, $A \leq\ (<)\ B$ means $a_{ij} \leq\ (<)\ b_{ij}$, for all i, j.

Let S be the signature matrix whose diagonal entries alternate in sign beginning with $+$. For $A, B \in \mathbb{R}^{m\times n}$, we write $A \overset{*}{\leq} B$ if and only if $SAS \leq SBS$ and $A \overset{*}{<} B$ if and only if $SAS < SBS$, and we call this the **"checkerboard" partial order** on real matrices.

Facts:

1. [Sch30] Let A be an $m \times n$ real matrix with $m \geq n$. If A is totally positive, then $V^+(A\mathbf{x}) \leq V^-(\mathbf{x})$, for all nonzero $\mathbf{x} \in \mathbb{R}^n$.
2. Let $A = [a_{ij}]$ be an $n \times n$ totally nonnegative matrix. Then:
   - Hadamard: [GK02, pp. 91–97], [Kot53]
     $\det A \leq \prod_{i=1}^n a_{ii}$,
   - Fischer: [GK02, pp. 91–97], [Kot53] Let $S \subseteq N = \{1, 2, \ldots, n\}$,
     $\det A \leq \det A[S] \cdot \det A[N \setminus S]$,
   - Koteljanskii: [Kot53] Let $S,\ T \subseteq N$,
     $\det A[S \cup T] \cdot \det A[S \cap T] \leq \det A[S] \cdot \det A[T]$.

   The above three determinantal inequalities also hold for positive semidefinite matrices. (See Chapter 8.5.)
3. [FGJ03] Let $\alpha_1, \alpha_2, \beta_1$, and β_2 be subsets of $\{1, 2, \ldots, n\}$, and let A be any TN matrix. Then

   $$\det A[\alpha_1] \det A[\alpha_2] \leq \det A[\beta_1] \det A[\beta_2]$$

   if and only if each index i has the same multiplicity in the multiset $\alpha_1 \cup \alpha_2$ and the multiset $\beta_1 \cup \beta_2$, and $\max(|\alpha_1 \cap L|, |\alpha_2 \cap L|) \geq \max(|\beta_1 \cap L|, |\beta_2 \cap L|)$, for every contiguous (i.e., $d(L) = 0$) $L \subseteq \{1, 2, \ldots, n\}$ (see [FGJ03] for other classes of principal-minor inequalities).
4. [FJS00] Let A be an $n \times n$ totally nonnegative matrix with $\det A(\{1\}) \neq 0$. Then $A - xE_{11}$ is totally nonnegative for all $x \in [0, \frac{\det A}{\det A(\{1\})}]$.
5. [FJ98] Let A be an $n \times n$ TN matrix partitioned as follows

   $$A = \begin{bmatrix} A_{11} & \mathbf{b} \\ \mathbf{c}^T & d \end{bmatrix},$$

   where A_{11} is $(n-1) \times (n-1)$. Suppose that $\operatorname{rank}(A_{11}) = p$. Then either $\operatorname{rank}([A_{11}, \mathbf{b}]) = p$ or $\operatorname{rank}[A_{11}^T, \mathbf{c}]^T = p$. See [FJ98] for other types of row and column inclusion results for TN matrices.
6. [CFJ01] Let T be an $n \times n$ totally nonnegative tridiagonal matrix. Then the Hadamard product of T with any other TN matrix is again TN.

7. [CFJ01][Mar70b] The Hadamard product of any two $n \times n$ tridiagonal totally nonnegative matrices is again totally nonnegative.

8. [CFJ01] Let $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$ be a 3×3 totally nonnegative matrix. Then A has the property that $A \circ B$ is TN for all B TN if and only if

$$aei + gbf \geq afh + dbi,$$
$$aei + dch \geq afh + dbi.$$

9. [CFJ01] Let A be an $n \times n$ totally nonnegative matrix with the property that $A \circ B$ is TN for all B TN, and suppose B is any $n \times n$ totally nonnegative matrix. Then

$$\det(A \circ B) \geq \det B \prod_{i=1}^{n} a_{ii}.$$

10. [Ste91] Let $A = [a_{ij}] \in \mathbb{R}^{n \times n}$, and let S_n denote the symmetric group on n symbols. If χ is an irreducible character of S_n, then the corresponding matrix function

$$[a_{ij}] \mapsto \sum_{\omega \in S_n} \chi(\omega) \prod_{i=1}^{n} a_{i,\omega(i)}$$

is called an *immanant*. For example, if χ is the trivial character $\chi(w) = 1$, then the corresponding immanant is the permanent and, if χ is $\text{sgn}(\omega)$, then the immanant is the determinant. Then every immanant of a totally nonnegative matrix is nonnegative.

11. [Gar96] If $A, B, C \in \mathbb{R}^{n \times n}$, $A \overset{*}{\leq} C \overset{*}{\leq} B$, and A and B are TP, then $\det C > 0$.
12. [Gar96] If $A, B, C \in \mathbb{R}^{n \times n}$, $A \overset{*}{\leq} C \overset{*}{\leq} B$, and A and B are TP, then C is TP.

Examples:

1. Let

$$P_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 1 & 3 & 6 & 10 \\ 1 & 4 & 10 & 20 \end{bmatrix} \quad \text{and} \quad \text{let } \mathbf{x} = \begin{bmatrix} 1 \\ -2 \\ -5 \\ 4 \end{bmatrix}.$$

Then $V^-(\mathbf{x}) = 2$ and $P_4\mathbf{x} = [-2, -2, 5, 23]^T$, so $V^+(P_4\mathbf{x}) = 1$. Hence, Schoenberg's variation diminishing property (Fact 1 of section 21.5) holds in this case.

2. Let

$$H = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix} \quad \text{and} \quad \text{let } \mathbf{x} = \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \end{bmatrix}.$$

Then $V^-(\mathbf{x}) = 2$ and $H\mathbf{x} = [0, -1, 0, 0]^T$, so $V^+(H\mathbf{x}) = 3$. Hence, Schoenberg's variation diminishing property (Fact 1 of section 21.5) does not hold in general for TN matrices.

3. If $0 < A < B$ and $A, B \in \mathbb{R}^{n\times n}$ are TP, then not all matrices in the interval between A and B need be TP. Let

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 4 & 5 \\ 5 & 7 \end{bmatrix}.$$

Then A, B are TP and

$$C = \begin{bmatrix} 3 & 4 \\ 4 & 5 \end{bmatrix}$$

satisfies $A < C < B$, and C is not TP.

4. If TP is replaced by TN, then Fact 12 no longer holds. For example,

$$\begin{bmatrix} 2 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix} \overset{*}{\leq} \begin{bmatrix} 3 & 0 & 4 \\ 0 & 0 & 0 \\ 4 & 0 & 5 \end{bmatrix} \overset{*}{\leq} \begin{bmatrix} 4 & 0 & 5 \\ 0 & 0 & 0 \\ 5 & 0 & 7 \end{bmatrix};$$

however, both end matrices are TN while the middle matrix is not.

5. A polynomial $f(x)$ is *stable* if all the zeros of $f(x)$ have negative real parts, which is equivalent to the the Routh–Hurwitz matrix (Example 4 of Section 21.1) formed from f being totally nonnegative [Asn70]. Suppose $f(x) = \sum_{i=0}^{n} a_i x^i$ and $g(x) = \sum_{i=0}^{m} b_i x^i$ are two polynomials of degree n and m, respectively. Then the *Hadamard product of f and g* is the polynomial $(f \circ g)(x) = \sum_{i=0}^{k} a_i b_i x^i$, where $k = \min(m, n)$. In [GW96a], it is proved that the Hadamard product of stable polynomials is stable. Hence, the Hadamard product of two totally nonnegative Routh–Hurwitz matrices is in turn a totally nonnegative matrix ([GW96a]). See also [GW96b] for a list of other subclasses of TN matrices that are closed under Hadamard multiplication.

6. Let $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ and let $B = A^T$. Then A and B are TN, $A \circ B = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$, and $\det(A \circ B) = -1 < 0$. Thus, $A \circ B$ is not TN.

7. (Polya matrix) Let $q \in (0, 1)$. The $n \times n$ Polya matrix Q has its $(i, j)^{th}$ entry equal to q^{-2ij}. Then Q is totally positive for all n (see [Whi52]). In fact, Q is diagonally equivalent to a TP Vandermonde matrix. Suppose Q represents the 3×3 Polya matrix. Then Q satisfies that $Q \circ A$ is TN for all A TN whenever $q \in (0, \sqrt{1/\mu})$, where $\mu = \frac{1+\sqrt{5}}{2}$ (the golden mean).

8. The bidiagonal factorization (21.2) from Fact 5 of section 21.2 for TN matrices was used in [Loe55] to show that for each nonsingular $n \times n$ TN matrix A there is a piece-wise continuous family of matrices $\Omega(t)$ of a special form such that the unique solution of the initial value problem

$$\frac{dA(t)}{dt} = \Omega(t)A(t), \; A(0) = I \tag{21.5}$$

has $A(1) = A$. Let $A(t) = [a_{ij}(t)]$ be a differentiable matrix-valued function of t such that $A(t)$ is nonsingular and TN for all $t \in [0, 1]$, and $A(0) = I$. Then

$$\Omega \equiv \left(\frac{dA(t)}{dt}\right)_{t=0} \tag{21.6}$$

is called an *infinitesimal element* of the semigroup of nonsingular TN matrices. By (21.2) every nonsingular TN matrix can be obtained from the solution of the initial value problem (21.5) in which all $\Omega(t)$ are infinitesimal elements.

9. If the main diagonal entries of a TP matrix are all ones, then it is not difficult to observe that as you move away from the diagonal there is a "drop off" effect in the entries of this matrix. Craven and

Csordas [CC98] have worked out an enticing sufficient "drop off" condition for a matrix to be TP. If $A = [a_{ij}]$ is an $n \times n$ matrix with positive entries and satisfies

$$a_{ij}a_{i+1,j+1} \geq c_0 a_{i,j+1}a_{i+1,j},$$

where $c_0 = 4.07959562349...$, then A is TP. This condition is particularly appealing for both Hankel and Toeplitz matrices. Recall that a Hankel matrix is an $(n+1) \times (n+1)$ matrix of the form

$$\begin{bmatrix} a_0 & a_1 & \cdots & a_n \\ a_1 & a_2 & \cdots & a_{n+1} \\ \vdots & \vdots & \cdots & \vdots \\ a_n & a_{n+1} & \cdots & a_{2n} \end{bmatrix}.$$

So, if the positive sequence $\{a_i\}$ satisfies $a_{k-1}a_{k+1} \geq c_0 a_k^2$, then the corresponding Hankel matrix is TP. An $(n+1) \times (n+1)$ Toeplitz matrix is of the form

$$\begin{bmatrix} a_0 & a_1 & a_2 & \cdots & a_n \\ a_{-1} & a_0 & a_1 & \cdots & a_{n-1} \\ a_{-2} & a_{-1} & a_0 & \cdots & a_{n-2} \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ a_{-n} & a_{-(n-1)} & a_{-(n-2)} & \cdots & a_0 \end{bmatrix}.$$

Hence, if the positive sequence $\{a_i\}$ satisfies $a_k^2 \geq c_0 a_{k-1}a_{k+1}$, then the corresponding Toeplitz matrix is TP.

10. A sequence $a_0, a_1, \ldots$ of real numbers is called *totally positive* if the two-way infinite matrix given by

$$\begin{bmatrix} a_0 & 0 & 0 & \cdots \\ a_1 & a_0 & 0 & \cdots \\ a_2 & a_1 & a_0 & \cdots \\ \vdots & \vdots & \vdots & \end{bmatrix}$$

is TP. As usual, an infinite matrix is TP all of its minors are positive. Notice that the above matrix is a Toeplitz matrix. Studying the functions that generate totally positive sequences was a difficult and important step in the area of totally positive matrices; $f(x)$ generates the sequence $a_0, a_1, \ldots$ if $f(x) = a_0 + a_1x + a_2x^2 + \cdots$. In Aissen et al. [ASW52] (see also [Edr52]), it was shown that the above two-way infinite Toeplitz matrix is TP (i.e., the corresponding sequence is totally positive) if and only if the generating function $f(x)$ for the sequence $a_0, a_1, \ldots$ has the form

$$f(x) = e^{\gamma x}\frac{\prod_{\nu=1}^{\infty}(1+\alpha_\nu x)}{\prod_{\nu=1}^{\infty}(1-\beta_\nu x)},$$

where $\gamma, \alpha_\nu, \beta_\nu \geq 0$, and $\sum \alpha_\nu$ and $\sum \beta_\nu$ are convergent.

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22
Linear Preserver Problems

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Linear preservers are linear maps on linear spaces of matrices that leave certain subsets, properties, relations, functions, etc., invariant. Linear preserver problems ask what is the general form of such maps. Describing the structure of such maps often gives a deeper understanding of the matrix sets, functions, or relations under the consideration. Some of the linear preserver problems are motivated by applications (system theory, quantum mechanics, etc.).

22.1 Basic Concepts

Definitions:

Let $\mathcal{V}$ be a linear subspace of $F^{m\times n}$. Let f be a (scalar-valued, vector-valued, or set-valued) function on $\mathcal{V}$, $\mathcal{M}$ a subset of $\mathcal{V}$, and $\sim$ a relation defined on $\mathcal{V}$.

A linear map $\phi : \mathcal{V} \to \mathcal{V}$ is called a **linear preserver of function** f if $f(\phi(A)) = f(A)$ for every $A \in \mathcal{V}$.

A linear map $\phi : \mathcal{V} \to \mathcal{V}$ **preserves** $\mathcal{M}$ if $\phi(\mathcal{M}) \subseteq \mathcal{M}$.

The map ϕ **strongly preserves** $\mathcal{M}$ if $\phi(\mathcal{M}) = \mathcal{M}$.

A linear map $\phi : \mathcal{V} \to \mathcal{V}$ **preserves the relation** $\sim$ if $\phi(A) \sim \phi(B)$ whenever $A \sim B$, $A, B \in \mathcal{V}$.

If for every pair $A, B \in \mathcal{V}$ we have $\phi(A) \sim \phi(B)$ if and only if $A \sim B$, then ϕ **strongly preserves the relation** $\sim$.

Facts:

1. If linear maps $\phi : \mathcal{V} \to \mathcal{V}$ and $\psi : \mathcal{V} \to \mathcal{V}$ both preserve $\mathcal{M}$, then $\phi\psi : \mathcal{V} \to \mathcal{V}$ preserves $\mathcal{M}$. Consequently, the set of all linear transformations on $\mathcal{V}$ preserving $\mathcal{M}$ is a multiplicative semigroup.
2. The set of all bijective linear transformations strongly preserving $\mathcal{M}$ is a multiplicative group.
3. [BLL92, p. 41] Let $\mathcal{M} \subseteq \mathcal{V}$ be an algebraic subset and $\phi : \mathcal{V} \to \mathcal{V}$ a bijective linear map satisfying $\phi(\mathcal{M}) \subseteq \mathcal{M}$. Then ϕ strongly preserves $\mathcal{M}$.

Examples:

1. Let $R \in F^{m\times m}$ and $S \in F^{n\times n}$. The linear map $\phi: F^{m\times n} \rightarrow F^{m\times n}$ defined by $\phi(A) = RAS$, $A \in F^{m\times n}$, preserves the set of all matrices of rank at most one. In general, such a map does not preserve this set strongly.
2. If R and S in the previous example are invertible, then ϕ is a bijective linear map strongly preserving matrices of rank one. In fact, such a map ϕ strongly preserves matrices of rank k, $k = 1, \ldots, \min\{m, n\}$.
3. If $m = n$ and $R, S \in F^{n\times n}$, then the linear map $A \mapsto RA^T S$, $A \in F^{n\times n}$, preserves matrices of rank at most one. If both R and S are invertible, then ϕ strongly preserves the set of all matrices of rank k, $1 \leq k \leq n$.
4. Assume that $m = n$. Let $R \in F^{n\times n}$ be an invertible matrix, c a nonzero scalar, and $f : F^{n\times n} \rightarrow F$ a linear functional. Then both maps $A \mapsto cRAR^{-1} + f(A)I$, $A \in F^{n\times n}$, and $A \mapsto cRA^T R^{-1} + f(A)I$, $A \in F^{n\times n}$, strongly preserve commutativity; that is, if $\phi : F^{n\times n} \rightarrow F^{n\times n}$ is any of these two maps, then for every pair $A, B \in F^{n\times n}$ we have $\phi(A)\phi(B) = \phi(B)\phi(A)$ if and only if $AB = BA$.
5. Let $\|\cdot\|$ be any norm on $\mathbb{C}^{m\times n}$. A linear map $\phi : \mathbb{C}^{m\times n} \rightarrow \mathbb{C}^{m\times n}$ is called an isometry if $\|\phi(A)\| = \|A\|$, $A \in \mathbb{C}^{m\times n}$. Thus, isometries are linear preservers of norm functions.
6. Let $1 \leq k < \min\{m, n\}$. A matrix $A \in F^{m\times n}$ is of rank at most k if the determinant of every $(k+1) \times (k+1)$ submatrix of A is zero. Thus, the set of all $m \times n$ matrices of rank at most k is an algebraic subset of $F^{m\times n}$.
7. The set of all nilpotent $n \times n$ matrices is an algebraic subset of $F^{n\times n}$. More generally, given a polynomial $p \in F[X]$, the set of all matrices $A \in F^{n\times n}$ satisfying $p(A) = 0$ is an algebraic set. Hence, bijective linear maps on $F^{n\times n}$ preserving idempotent matrices, nilpotents, involutions, etc. preserve these sets strongly.

22.2 Standard Forms

Definitions:

Let $\mathcal{V}$ be a linear subspace of $F^{m\times n}$ and let P be a preserving property which makes sense for linear maps acting on $\mathcal{V}$ (P may be the property of preserving a certain subset of $\mathcal{V}$ or the property of preserving a certain relation on $\mathcal{V}$ or the property of preserving a certain function defined on $\mathcal{V}$). A **linear preserver problem** corresponding to the property P is the problem of characterizing all linear (bijective) maps on $\mathcal{V}$ satisfying this property. Very often, linear preservers have the standard forms (see the next section). Occasionally, there are interesting exceptional cases especially in low dimensions (see later examples).

Let $R \in F^{m\times m}$ and $S \in F^{n\times n}$ be invertible matrices. A map $\phi : F^{m\times n} \rightarrow F^{m\times n}$ is called an (R, S)-**standard map** if either $\phi(A) = RAS$, $A \in F^{m\times n}$, or $m = n$ and $\phi(A) = RA^T S$, $A \in F^{n\times n}$. In many cases we assume that R and S satisfy some additional assumptions.

Let $R \in F^{n\times n}$ be an invertible matrix, c a nonzero scalar, and $f : F^{n\times n} \rightarrow F$ a linear functional. A map $\phi : F^{n\times n} \rightarrow F^{n\times n}$ is called an (R, c, f)-**standard map** if either $\phi(A) = cRAR^{-1} + f(A)I$, $A \in F^{n\times n}$, or $\phi(A) = cRA^T R^{-1} + f(A)I$, $A \in F^{n\times n}$.

When we consider linear preservers on proper subspaces $\mathcal{V} \subset F^{m\times n}$, we usually have to modify the notion of standard maps. Let us consider the case when $\mathcal{V} = \mathcal{T}_n \subset F^{n\times n}$ is the subalgebra of all upper triangular matrices. The **flip map** $A \mapsto A^f$, $A \in \mathcal{T}_n$, is defined as the transposition over the antidiagonal; that is, $A^f = GA^T G$, where $G = E_{1n} + E_{2,n-1} + \ldots + E_{n1}$ (see Example 1). Standard maps on $\mathcal{T}_n$ are maps of the form

$$A \mapsto RAS, \quad A \in \mathcal{T}_n,$$

$$A \mapsto RA^f S, \quad A \in \mathcal{T}_n,$$

$$A \mapsto cRAR^{-1} + f(A)I, \quad A \in \mathcal{T}_n,$$

and

$$A \mapsto cRA^f R^{-1} + f(A)I, \quad A \in \mathcal{T}_n,$$

where R and S are invertible upper triangular matrices, c is a nonzero scalar, and f is a linear functional on $\mathcal{T}_n$.

Facts:

1. Every (R, S)-standard map is bijective. It strongly preserves matrices of rank k, $k = 1, \ldots, \min\{m, n\}$. Every (R, c, f)-standard map is either bijective, or its kernel is the one-dimensional subspace consisting of all scalar matrices.
2. Let $U \in \mathbb{C}^{m\times m}$ and $V \in \mathbb{C}^{n\times n}$ be unitary matrices. Then a (U, V)-standard map preserves singular values and, hence, all functions of singular values including unitarily invariant norms.
3. Let $R \in \mathbb{C}^{n\times n}$ be an invertible matrix. Then an (R, R^{-1})-standard map on $\mathbb{C}^{n\times n}$ preserves spectrum, idempotents, nilpotents, similarity, zero products, etc.
4. If $U \in \mathbb{C}^{n\times n}$ is a unitary matrix, then a (U, U^*)-standard map defined on $\mathbb{C}^{n\times n}$ strongly preserves the set of all orthogonal idempotents, the set of all normal matrices, numerical range, etc.
5. If $A, B \in \mathcal{T}_n$, then $(AB)^f = B^f A^f$.

Examples:

1.
$$\begin{bmatrix} a & b & c & d \\ 0 & e & f & g \\ 0 & 0 & h & i \\ 0 & 0 & 0 & j \end{bmatrix}^f = \begin{bmatrix} j & i & g & d \\ 0 & h & f & c \\ 0 & 0 & e & b \\ 0 & 0 & 0 & a \end{bmatrix}.$$

2. A map $\phi : \mathbb{C}^{2\times 2} \to \mathbb{C}^{2\times 2}$ given by
$$\phi\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = \begin{bmatrix} a & -b \\ c & d \end{bmatrix}$$
is a bijective linear map that strongly preserves commutativity but is not of a standard form. More generally, any bijective linear map $\phi : \mathbb{C}^{2\times 2} \to \mathbb{C}^{2\times 2}$ satisfying $\phi(I) = I$ strongly preserves commutativity [Kun99]. In higher dimensions there are no nonstandard bijective linear maps preserving commutativity [BLL92, p. 76].
3. Let $\mathcal{W} \subset \mathbb{C}^{n\times n}$ be any linear subspace of matrices with the property that $AB = BA$ for every pair $A, B \in \mathcal{W}$. Assume that $\phi : \mathbb{C}^{n\times n} \to \mathbb{C}^{n\times n}$ is a linear map whose range is contained in $\mathcal{W}$. Then ϕ is a nonstandard map preserving commutativity. If $n > 1$, then it does not preserve commutativity strongly. A map $\phi : \mathbb{C}^{2\times 2} \to \mathbb{C}^{2\times 2}$ defined by
$$\phi\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = \begin{bmatrix} a & b \\ -b & a \end{bmatrix}$$
is a concrete example of such map.
4. A map $\phi : \mathbb{R}^{4\times 4} \to \mathbb{R}^{4\times 4}$ given by
$$\phi\left(\begin{bmatrix} a & b & c & d \\ * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{bmatrix}\right) = \begin{bmatrix} a & b & c & d \\ -b & a & -d & c \\ -c & d & a & -b \\ -d & -c & b & a \end{bmatrix}$$

preserves the real orthogonal group, that is, $\phi(O)$ is an orthogonal matrix for every orthogonal matrix $O \in \mathbb{R}^{4\times 4}$. Note that the right-hand side of the above equation is the standard matrix representation of the quaternion $a+bi+cj+dk$. Similar constructions with matrix representations of complex and Cayley numbers give nonstandard linear preservers of real orthogonal group on $\mathbb{R}^{2\times 2}$ and $\mathbb{R}^{8\times 8}$. If $\phi : \mathbb{R}^{n\times n} \to \mathbb{R}^{n\times n}$ is a linear preserver of orthogonal group and $n \notin \{2, 4, 8\}$, then ϕ is a (U, V)-standard map with U and V being orthogonal matrices [LP01, p. 601].

5. A linear map ϕ acting on 4×4 upper triangular matrices given by

$$\begin{bmatrix} a & b & c & d \\ 0 & e & f & g \\ 0 & 0 & h & i \\ 0 & 0 & 0 & j \end{bmatrix} \mapsto \begin{bmatrix} e & f & g & b \\ 0 & j & i & c \\ 0 & 0 & h & d \\ 0 & 0 & 0 & a \end{bmatrix}$$

is a nonstandard bijective linear map strongly preserving the set of invertible matrices. All that is important in this example is that A and $\phi(A)$ have the same diagonal entries up to a permutation.

22.3 Standard Linear Preserver Problems

Definitions:

Linear preserver problems ask what is the general form of (bijective) linear maps on matrix spaces having a certain preserving property. When the general form of linear preservers under the consideration is one of the standard forms, we speak of a **standard linear preserver problem**. The following list of some most important standard linear preserver results is far from being complete. Many more results can be found in the survey [BLL92].

Facts:

1. [BLL92, Theorem 2.2], [LT92, Prop. 3] Let m, n be positive integers and $\phi : \mathbb{C}^{m\times n} \to \mathbb{C}^{m\times n}$ a linear map. Assume that one of the following two conditions is satisfied:
   - Let k be a positive integer, $k \le \min\{m, n\}$, and assume that $\operatorname{rank}\phi(A) = k$ whenever $\operatorname{rank} A = k$.
   - ϕ is invertible and $\operatorname{rank}\phi(A) = \operatorname{rank}\phi(B)$ whenever $\operatorname{rank} A = \operatorname{rank} B$.

   Then ϕ is an (R, S)-standard map for some invertible matrices $R \in \mathbb{C}^{m\times m}$ and $S \in \mathbb{C}^{n\times n}$.
2. [BLL92, p. 9] Let F be a field with more than three elements, k a positive integer, $k \le \min\{m, n\}$, and $\phi : F^{m\times n} \to F^{m\times n}$ an invertible linear map such that $\operatorname{rank}\phi(A) = k$ whenever $\operatorname{rank} A = k$. Then ϕ is an (R, S)-standard map for some invertible matrices $R \in F^{m\times m}$ and $S \in F^{n\times n}$.
3. [BLL92, Theorem 2.6] Let $\phi : F^{m\times n} \to F^{p\times q}$ be a linear map such that $\operatorname{rank}\phi(A) \le 1$ whenever $\operatorname{rank} A = 1$. Then either

   (a) $\phi(A) = RAS$ for some $R \in F^{p\times m}$ and some $S \in F^{n\times q}$ or
   (b) $\phi(A) = RA^T S$ for some $R \in F^{p\times n}$ and some $S \in F^{m\times q}$ or
   (c) The range of ϕ is contained in the set of all matrices of rank at most one.

4. [BLL92, p. 10] Let F be an infinite field with characteristic $\neq 2$, $\mathcal{S}_n$ the space of all $n \times n$ symmetric matrices over F, and let k be an integer, $1 \le k \le n$. If $\phi : \mathcal{S}_n \to \mathcal{S}_n$ is an invertible linear rank k preserver, then $\phi(A) = cRAR^T$ for every $A \in \mathcal{S}$. Here, c is a nonzero scalar and R an invertible $n \times n$ matrix.
5. [BLL92, Theorem 2.9] Let F be a field with characteristic $\neq 2$. Assume that F has more than three elements. Let $\phi : \mathcal{S}_n \to \mathcal{S}_m$ be a linear map such that $\operatorname{rank}\phi(A) \le 1$ whenever $\operatorname{rank} A = 1$. Then either

(a) There exist an $m \times n$ matrix R and a scalar c such that $\phi(A) = cRAR^T$ or

(b) The range of ϕ is contained in a linear span of some rank one $m \times m$ symmetric matrix.

6. [BLL92, Theorem 2.7, Theorem 2.8, p. 25] Let ϕ be a real linear map acting on the real linear space $\mathcal{H}_n$ of all $n \times n$ Hermitian complex matrices. Let π, ν, δ be nonnegative integers with $\pi + \nu + \delta = n$ and denote by $G(\pi, \nu, \delta)$ the set of all $A \in \mathcal{H}_n$ with inertia $\text{in}(A) = (\pi, \nu, \delta)$. Assume that one of the following is satisfied:

   - $n \geq 3$, $k < n$, ϕ is invertible, and $\text{rank}\,\phi(A) \leq k$ whenever $\text{rank}\,A = k$.
   - $n \geq 2$, the range of ϕ is not one-dimensional, and $\text{rank}\,\phi(A) = 1$ whenever $\text{rank}\,A = 1$.
   - The triple (π, ν, δ) does not belong to the set $\{(n, 0, 0), (0, n, 0), (0, 0, n)\}$, ϕ is bijective, and $\phi(G(\pi, \nu, \delta)) \subseteq G(\pi, \nu, \delta)$.

   Then there exist an invertible $n \times n$ complex matrix R and a constant $c \in \{-1, 1\}$ such that either

   (a) $\phi(A) = cRAR^*$ for every $A \in \mathcal{H}_n$ or

   (b) $\phi(A) = cRA^T R^*$ for every $A \in \mathcal{H}_n$.

   Of course, if the third of the above conditions is satisfied, then the possibility $c = -1$ can occur only when $\pi = \nu$.
7. [BLL92, p. 76] Let $n \geq 3$ and let $\phi : F^{n \times n} \to F^{n \times n}$ be an invertible linear map such that $\phi(A)\phi(B) = \phi(B)\phi(A)$ whenever A and B commute. Then ϕ is an (R, c, f)-standard map for some invertible $R \in F^{n \times n}$, some nonzero scalar c, and some linear functional $f : F^{n \times n} \to F$.
8. [LP01, Theorem 2.3] Let $\phi : \mathbb{C}^{n \times n} \to \mathbb{C}^{n \times n}$ be a linear similarity preserving map. Then either

   (a) ϕ is an (R, c, f)-standard map or

   (b) $\phi(A) = (\text{tr}\,A)B$, $A \in \mathbb{C}^{n \times n}$.

   Here, $B, R \in \mathbb{C}^{n \times n}$ and R is invertible, c is a nonzero complex number, and $f : \mathbb{C}^{n \times n} \to \mathbb{C}$ is a functional of the form $f(A) = b\text{tr}\,A$, $A \in \mathbb{C}^{n \times n}$, for some $b \in \mathbb{C}$.
9. [BLL92, p. 77] Let ϕ be a real-linear unitary similarity preserving map on $\mathcal{H}_n$. Then either

   (a) $\phi(A) = (\text{tr}\,A)B$, $A \in \mathcal{H}_n$ or

   (b) $\phi(A) = cUAU^* + b(\text{tr}\,A)I$, $A \in \mathcal{H}_n$ or

   (c) $\phi(A) = cUA^T U^* + b(\text{tr}\,A)I$, $A \in \mathcal{H}_n$.

   Here, B is a Hermitian matrix, U is a unitary matrix, and b, c are real constants.
10. [BLL92, Theorem 4.7.6] Let $n > 2$ and let F be an algebraically closed field of characteristic zero. Let p be a polynomial of degree n with at least two distinct roots. Let us write p as $p(x) = x^k q(x)$ with $k \geq 0$ and $q(0) \neq 0$. Assume that $\phi : F^{n \times n} \to F^{n \times n}$ is an invertible linear map preserving the set of all matrices annihilated by p. Then either

    (a) $\phi(A) = cRAR^{-1}$, $A \in F^{n \times n}$ or

    (b) $\phi(A) = cRA^T R^{-1}$, $A \in F^{n \times n}$.

    Here, R is an invertible matrix and c is a constant permuting the roots of q; that is, $q(c\lambda) = 0$ for each $\lambda \in F$ satisfying $q(\lambda) = 0$.
11. [BLL92, p. 48] Let $sl_n \subset F^{n \times n}$ be the linear space of all trace zero matrices and $\phi : sl_n \to sl_n$ an invertible linear map preserving the set of all nilpotent matrices. Then there exist an invertible matrix $R \in F^{n \times n}$ and a nonzero scalar c such that either

    (a) $\phi(A) = cRAR^{-1}$, $A \in sl_n$ or

    (b) $\phi(A) = cRA^T R^{-1}$, $A \in sl_n$.

    When considering linear preservers of nilpotent matrices one should observe first that the linear span of all nilpotent matrices is sl_n and, therefore, it is natural to confine maps under consideration to this subspace of codimension one.

12. [GLS00, pp. 76, 78] Let F be an algebraically closed field of characteristic 0, m, n positive integers, and $\phi : F^{n\times n} \to F^{m\times m}$ a linear transformation. If ϕ is nonzero and maps idempotent matrices to idempotent matrices, then $m \geq n$ and there exist an invertible matrix $R \in F^{m\times m}$ and nonnegative integers k_1, k_2 such that $1 \leq k_1 + k_2$, $(k_1 + k_2)n \leq m$ and

$$\phi(A) = R(A \oplus \ldots \oplus A \oplus A^T \oplus \ldots \oplus A^T \oplus 0)R^{-1}$$

for every $A \in F^{n\times n}$. In the above block diagonal direct sum the matrix A appears k_1 times, A^T appears k_2 times, and 0 is the zero matrix of the appropriate size (possibly absent). If $p \in F[X]$ is a polynomial of degree > 1 with simple zeros (each zero has multiplicity one), ϕ is unital and maps every $A \in F^{n\times n}$ satisfying $p(A) = 0$ into some $m \times m$ matrix annihilated by p, then ϕ is of the above described form with $(k_1 + k_2)n = m$.
13. [BLL92, Theorem 4.6.2] Let $\phi : \mathbb{C}^{n\times n} \to \mathbb{C}^{n\times n}$ be a linear map preserving the unitary group. Then ϕ is a (U, V)-standard map for some unitary matrices $U, V \in \mathbb{C}^{n\times n}$.
14. [KH92] Let $\phi : \mathbb{C}^{n\times n} \to \mathbb{C}^{n\times n}$ be a linear map preserving normal matrices. Then either

    (a) $\phi(A) = cUAU^* + f(A)I$, $A \in \mathbb{C}^{n\times n}$ or
    (b) $\phi(A) = cUA^tU^* + f(A)I$, $A \in \mathbb{C}^{n\times n}$ or
    (c) the range of ϕ is contained in the set of normal matrices.

    Here, U is a unitary matrix, c is a nonzero scalar, and f is a linear functional on $\mathbb{C}^{n\times n}$.
15. [LP01, p. 595] Let $\|\cdot\|$ be a unitarily invariant norm on $\mathbb{C}^{m\times n}$ that is not a multiple of the Frobenius norm defined by $\|A\| = \sqrt{\operatorname{tr}(AA^*)}$. The group of linear preservers of $\|\cdot\|$ on $\mathbb{C}^{m\times n}$ is the group of all (U, V)-standard maps, where $U \in \mathbb{C}^{m\times m}$ and $V \in \mathbb{C}^{n\times n}$ are unitary matrices. Of course, if $\|\cdot\|$ is a mulitple of the Frobenious norm, then the group of linear preservers of $\|\cdot\|$ on $\mathbb{C}^{m\times n}$ is the group of all unitary operators, i.e., those linear operators $\phi : \mathbb{C}^{m\times n} \to \mathbb{C}^{m\times n}$ that preserve the usual inner product $\langle A, B\rangle = \operatorname{tr}(AB^*)$ on $\mathbb{C}^{m\times n}$.
16. [BLL92, p. 63–64] Let $\phi : \mathbb{C}^{n\times n} \to \mathbb{C}^{n\times n}$ be a linear map preserving the numerical radius. Then either

    (a) $\phi(A) = cUAU^*$, $A \in \mathbb{C}^{n\times n}$ or
    (b) $\phi(A) = cUA^TU^*$, $A \in \mathbb{C}^{n\times n}$.

    Here, U is a unitary matrix and c a complex constant with $|c| = 1$.
17. [BLL92, Theorem 4.3.1] Let $n > 2$ and let $\phi : F^{n\times n} \to F^{n\times n}$ be a linear map preserving the permanent. Then ϕ is an (R, S)-standard map, where R and S are each a product of a diagonal and a permutation matrix, and the product of the two diagonal matrices has determinant one.
18. [CL98] Let $\phi : \mathcal{T}_n \to \mathcal{T}_n$ be a linear rank one preserver. Then either

    (a) The range of ϕ is the space of all matrices of the form

    $$\begin{bmatrix} * & * & \ldots & * \\ 0 & 0 & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & 0 \end{bmatrix}$$

    or

    (b) The range of ϕ is the space of all matrices of the form

    $$\begin{bmatrix} 0 & 0 & \ldots & 0 & * \\ 0 & 0 & \ldots & 0 & * \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \ldots & 0 & * \end{bmatrix}$$

    or

(c) $\phi(A) = RAS$ for some invertible $R, S \in \mathcal{T}_n$ or

(d) $\phi(A) = RA^f S$ for some invertible $R, S \in \mathcal{T}_n$.

Examples:

1. Let $n \geq 2$. Then the linear map $\phi : \mathcal{T}_n \to \mathcal{T}_n$ defined by

$$\phi\left(\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ 0 & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & a_{nn} \end{bmatrix}\right)$$

$$= \begin{bmatrix} a_{11} + a_{22} + \ldots + a_{nn} & a_{12} + a_{23} + \ldots + a_{n-1,n} & \dots & a_{1n} \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 \end{bmatrix}$$

is an example of a singular preserver of rank one.

2. The most important example of a nonstandard linear preserver problem is the problem of characterizing linear maps on $n \times n$ real or complex matrices preserving the set of positive semidefinite matrices. Let $R_1, \ldots, R_r, S_1, \ldots, S_k$ be $n \times n$ matrices. Then the linear map ϕ given by $\phi(A) = R_1 A R_1^* + \cdots + R_r A R_r^* + S_1 A^T S_1^* + \cdots + S_k A^T S_k^*$ is a linear preserver of positive semidefinite matrices. Such a map is called decomposable. In general it cannot be reduced to a single congruence or a single congruence composed with the transposition. Moreover, there exist linear maps on the space of $n \times n$ matrices preserving positive semidefinite matrices that are not decomposable. There is no general structural result for such maps.

22.4 Additive, Multiplicative, and Nonlinear Preservers

Definitions:

A map $\phi : F^{m\times n} \to F^{m\times n}$ is **additive** if $\phi(A + B) = \phi(A) + \phi(B)$, $A, B \in F^{m\times n}$. An additive map $\phi : F^{m\times n} \to F^{m\times n}$ having a certain preserving property is called an **additive preserver**.

A map $\phi : F^{n\times n} \to F^{n\times n}$ is **multiplicative** if $\phi(AB) = \phi(A)\phi(B)$, $A, B \in F^{n\times n}$. A multiplicative map $\phi : F^{n\times n} \to F^{n\times n}$ having a certain preserving property is called a **multiplicative preserver**.

Two matrices $A, B \in F^{m\times n}$ are said to be **adjacent** if $\text{rank}\,(A - B) = 1$.

A map $\phi : F^{n\times n} \to F^{n\times n}$ is called a **local similarity** if for every $A \in F^{n\times n}$ there exists an invertible $R_A \in F^{n\times n}$ such that $\phi(A) = R_A A R_A^{-1}$.

Let $f : F \to F$ be an automorphism of the field F. A map $\phi : F^{n\times n} \to F^{n\times n}$ defined by $\phi([a_{ij}]) = [f(a_{ij})]$ is called a **ring automorphism of** $F^{n\times n}$ **induced by** f.

Facts:

1. [BS00] Let $n \geq 2$ and assume that $\phi : F^{n\times n} \to F^{n\times n}$ is a surjective additive map preserving rank one matrices. Then there exist a pair of invertible matrices $R, S \in F^{n\times n}$ and an automorphism f of the field F such that ϕ is a composition of an (R, S)-standard map and a ring automorphism of $F^{n\times n}$ induced by f.
2. [GLR03] Let $SL(n, \mathbb{C})$ denote the group of all $n \times n$ complex matrices A such that $\det A = 1$. A multiplicative map $\phi : SL(n, \mathbb{C}) \to \mathbb{C}^{n\times n}$ satisfies $\rho(\phi(A)) = \rho(A)$ for every $A \in SL(n, \mathbb{C})$ if and only if there exists $S \in SL(n, \mathbb{C})$ such that either